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REVIEW & ANALYSIS

Biochar as a negative emission technology: A synthesis of field research on greenhouse gas emissions

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Abstract

Biochar is one of the few nature-based technologies with potential to help achieve net-zero emissions agriculture. Such an outcome would involve the mitigation of greenhouse gas (GHG) emission from agroecosystems and optimization of soil organic carbon sequestration. Interest in biochar application is heightened by its several co-benefits. Several reviews summarized past investigations on biochar, but these reviews mostly included laboratory, greenhouse, and mesocosm experiments. A synthesis of field studies is lacking, especially from a climate change mitigation standpoint. Our objectives are to (1) synthesize advances in field-based studies that have examined the GHG mitigation capacity of soil application of biochar and (2) identify limitations of the technology and research priorities. Field studies, published before 2022, were reviewed. Biochar has variable effects on GHG emissions, ranging from decrease, increase, to no change. Across studies, biochar reduced emissions of nitrous oxide (N_2O) by 18% and methane (CH_4) by 3% but increased carbon dioxide (CO_2) by 1.9%. When biochar was combined with N-fertilizer, it reduced CO_2, CH_4, and N_2O emissions in 61%, 64%, and 84% of the observations, and biochar plus other amendments reduced emissions in 78%, 92%, and 85% of the observations, respectively. Biochar has shown potential to reduce GHG emissions from soils, but long-term studies are needed to address discrepancies in emissions and identify best practices (rate, depth, and frequency) of biochar application to agricultural soils.

1 | INTRODUCTION

Global greenhouse gas [GHG, carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O)] emissions have risen by about 5.6%, 4.2%, and 2.7%, respectively, during the 2011–2020 decade (NOAA, 2022). In 2019, global GHG emissions amounted to 57.4 Gt CO_2 equivalent (Olivier & Peters, 2020) with agriculture (livestock and manure, agriculture soils, cropland, including rice cultivation) accounting for 16% of these emissions (Ritchie & Roser, 2022). These emissions could lead to irreversible changes to the global climate system (Platzer & Sarigul-Klijn, 2021). The time to implement

Abbreviations: C, carbon; CH_4, methane; CO_2, carbon dioxide; GHG, greenhouse gas; GWP, global warming potential; N, nitrogen; N_2O, nitrous oxide; NH_4^+, ammonium; NO_3^-, nitrate; SOC, soil organic carbon.

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solutions to address the threat of climate change is now (Amonette et al., 2021). The Paris Climate agreement of 2015 called for a global effort to limit the average temperature increase to 1.5˚C above preindustrial levels (IPCC, 2018; UNFCCC, 2015). To achieve this target, best management practices (BMPs) must be implemented to create net-negative emissions (GHG removal/uptake » emission) (Lal, 2021).

The application of biochar to agroecosystems is one of the proposed BMPs for achieving negative emission goals (Smith, 2016; Tisserant & Cherubini, 2019). Biochar technology can lead to C negative agricultural systems and help mitigate climate change (Goswami et al., 2021; Ong et al., 2021). Biochar is a carbon (C)-rich product obtained from pyrolysis (heating under oxygen-deficient conditions) of biomass (such as wood chips, crop residue, or manure) and used as a soil amendment or a C sequestration agent (Lehmann & Joseph, 2015; Mishra et al., 2022; Wang & Wang, 2019). Globally, it has been estimated that biochar use could reduce GHG emissions from soils by 3.4–6.3 Pg CO_2 equivalent (Lehmann et al., 2021). Conversion of biomass to biochar through pyrolysis protects the majority of biomass C from microbial decomposition. During the biochar manufacturing process, about 50% of the original C in the biomass is retained in the final product (Hammond et al., 2011), and the process also induces a reversal of the C cycle by removing organic compounds from the active C pool and converting them into refractory organic components (Das et al., 2014; Lee et al., 2010). It is estimated that, when biochar is incorporated into soil, it leads to sequestration of about 50% of the initial biomass C compared to the low amounts retained after burning (3%) or biological decomposition (<10%–20% after 5–10 years) (Lehmann et al., 2006). Thus, pyrolysis of biomass into biochar and subsequent application of biochar to soils can help build soil organic carbon (SOC) stocks and mitigate GHG emissions—two essential pathways toward net-negative emission agroecosystems and thus offset the global warming impacts of crop production (Figure 1).

Biochar differs from charcoal, mainly used as cooking and heating fuel, in that it is produced through pyrolysis and is applied as soil amendment among other uses. The term "biochar" has seen an exponential increase in use in the scientific literature over the last two decades (Figure 2), demonstrating increased interest of the scientific community on biochar for agricultural production and other purposes in several countries (Figure 3). This interest can be ascribed to reported beneficial impacts of biochar to: (1) long-term sequestration of atmospheric CO_2 in soils (Purakayastha et al., 2021), (2) improvement in soil health (Alkharabsheh et al., 2021; Amoah-Antwi et al., 2020; Darmawan et al., 2021) and fertility (Diatta et al., 2020; Mohawesh et al., 2021), (3) increase in soil water and nutrient retention (Are, 2019; Fischer et al., 2019; Razzaghi et al., 2020), (4) reduction in nitrogen (N) loss through leaching (Major et al., 2009), (5) enhancement of biological N-fixation in acidic soils (Ma, Egamberdieva, et al., 2019; Van Zwieten et al., 2015), (6) increase in crop yield (Farhangi-Abriz et al., 2021; Haque et al., 2019; Safaei et al., 2020), (7) alleviation of environmental pollution through soil and water remediation (Borchard et al., 2019; Li et al., 2020; Man et al., 2021), (8) restoration of degraded soils (Amoah-Antwi et al., 2020; Ghosh & Maiti, 2021; Rodriguez-Franco & Page-Dumroese, 2021), and (9) mitigation of climate change through GHG emission reduction and C sequestration in soils (Demir, 2020; Gupta et al., 2020: Purakayastha et al., 2021). A recent life cycle analysis has shown that C sequestration and GHG mitigation benefits of biochar overcompensate for CO_2, N_2O, and CH_4 emitted during biochar production (Matustik et al., 2020).

Core Ideas

- Biochar has variable effects on greenhouse gas emissions, ranging from decrease, increase, to no change.
- Across studies, biochar reduces emissions of nitrous oxide by −18% (−79 to 56) and methane by −3% (−358 to 400).
- Increases carbon dioxide emissions by 1.9 (−31 to 49) compared to control.
- Biochar is more effective when combined with mineral N-fertilizer or organic amendments (manure, straw).

Application of biochar to agricultural soils has shown variable effects on GHG emissions, ranging from decrease, increase, or no change. Beneficial effects of biochar depend on biochar properties that can vary greatly with type of feedstock (Abagandura et al., 2019; Fungo et al., 2014; He et al., 2017) and production process (Chun et al., 2021; Freitas et al., 2020; He et al., 2017). The benefits of biochar also vary with crop (Olszyk et al., 2020; Shakoor, Arif, et al., 2021), soil type (He et al., 2017; Kavitha et al., 2018), application rate (Guo, 2020), and climatic conditions (Blanco-Canqui, 2021; Lu et al., 2020).

Agricultural management practices (such as fertilizer and manure application and soil management) can stimulate GHG emissions (O'Connor et al., 2020; Panchasara et al., 2021). However, negative emissions in agroecosystems can be achieved by reducing the strength of GHG emissions (sources) and enhancing soil C pools (sink) (Lal, 2021). Biochar has been proposed as a soil amendment, with potential for not only increasing crop yield (Bai et al., 2022; Farhangi-Abriz et al., 2021) due to improvement in soil physicochemical and biological properties (Egamberdieva et al., 2022; Kapoor et al., 2022) but also for reducing GHG emissions from

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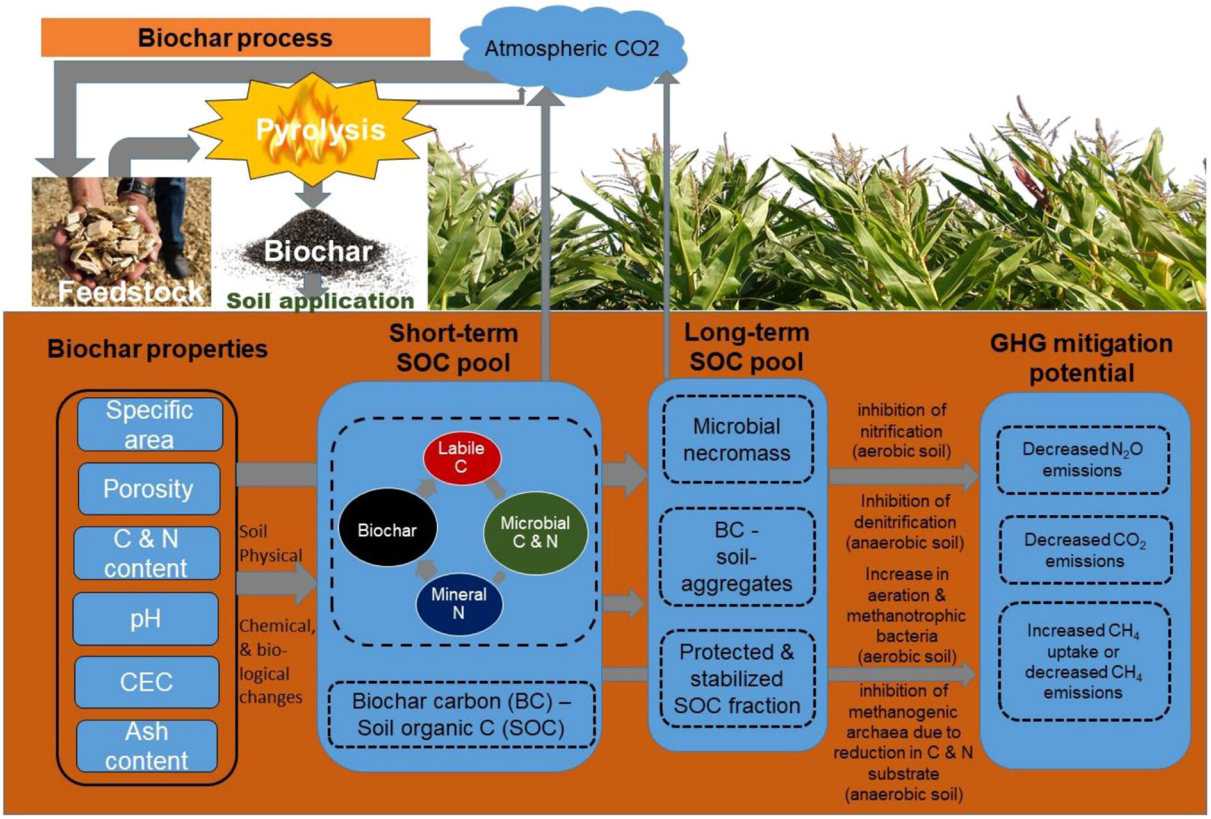


FIGURE 1 Conceptual diagram showing biochar process, properties, and impact on soil organic carbon pools and greenhouse gas emission mitigation in biochar-incorporated fields (Chagas et al., 2022; El-Naggar et al., 2018; Jiang et al., 2020; Liang et al., 2019; Sun et al., 2021; Van Zwieten et al., 2017; Yang et al., 2020; Zhang et al., 2019; Zhao et al., 2021).

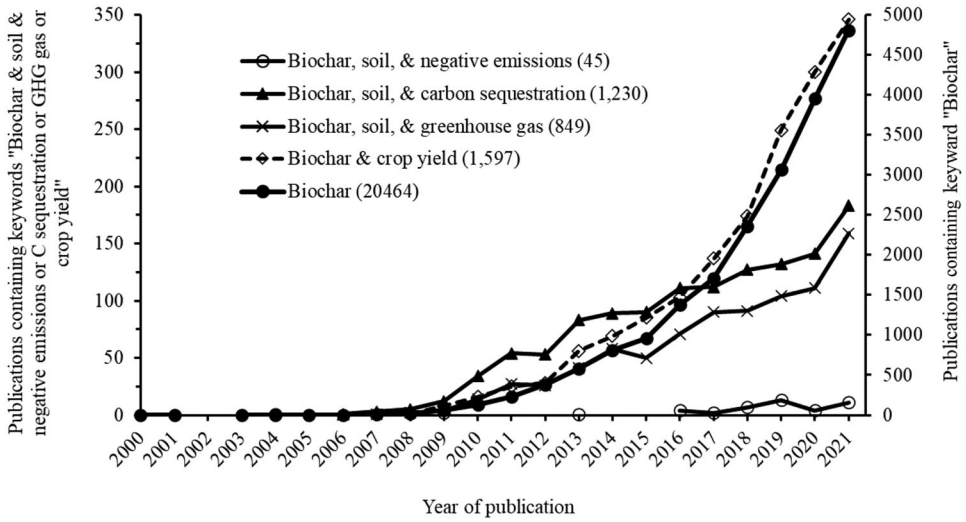


FIGURE 2 Number of published journal articles (in English) related to the impact of biochar on soil carbon sequestration, greenhouse gas emissions, negative emissions, and crop yield. *Source*: Scopus database (http://www.scopus.com/). Assessed 02/16/2022. Values in parenthesis are total number of publications published between 2000 and 2021.

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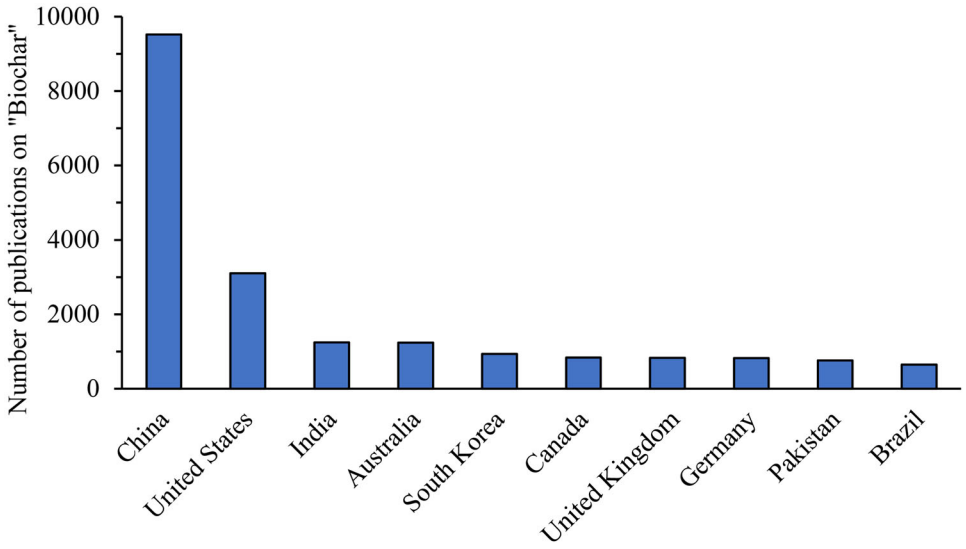


FIGURE 3 Number of publications on "Biochar" (in English) from top 10 countries. *Source*: Scopus database (https://www.scopus.com/). Accessed 12/29/2021.

agroecosystems. Overall, the climate change mitigation benefits of biochar stem from: (1) increased retention of C input through evolution of condensed aromatic structures that are highly stable and resistant to decomposition (Chagas et al., 2022; Gross et al., 2021) and (2) reduced GHG emissions (Gross et al., 2022; Shakoor, Arif, et al., 2021).

Further climate benefits of biochar can come from avoided GHG emissions from crop residues that would have been burned (Dhaliwal et al., 2022; Ravindra et al., 2019) or left on the field to decompose (Song et al., 2022). Emission avoidance can also be achieved when carbonaceous residues are converted to biochar instead of compost or manure that would otherwise be stockpiled resulting in increased CH_4 emissions (Agyarko-Mintah et al., 2017; Wu et al., 2017). Yet, biochar is not commonly used in agricultural production systems due to its high cost (Bach et al., 2016; Li et al., 2015). For example, the current price of biochar ranges from $1000 to $3300 Mg^{-1} (Farm Energy, 2019; Oregon Biochar Solutions, 2022; Wakefield Biochar, 2021) which, combined with a lack of proven agronomic and environmental benefits at large scale (Latawiec et al., 2017; Palai, 2020), has limited its use at the farm-scale. Although biochar has not yet been recognized as a method for generating C offsets, its potential should not be ignored because of its long residence time in soils, ranging from decades to thousands of years (Jiang et al., 2020; Kuzyakov et al., 2009; Major et al., 2010). Therefore, biochar application can be a valuable BMP to build SOC stocks (Chages et al., 2022; Seki et al., 2022) and reduce GHG emissions (Shakoor, Arif, et al., 2021; Stavi, 2012). However, to fully assess biochar as a potential C negative technology, its GHG emission mitigation benefits should be accounted for in addition to SOC sequestration (Terra Global Capital, 2010; The Climate Trust, 2010).

Due to evolving understanding of the diverse benefits of biochar and resulting increase in interest, extensive reviews and meta-analyses have been published, especially on crop growth and yield (Bai et al., 2022; Dai et al., 2020), soil health (Farhangi-Abriz et al., 2021; Kapoor et al., 2022), waste management and environmental remediation (Liu et al., 2022; Sun et al., 2021), and climate change mitigation (Lehmann et al., 2021; Shakoor, Arif, et al., 2021) (Figure 4). There has been a wealth of biochar-related studies conducted under laboratory, greenhouse, mesocosm, and field conditions, and available reviews have reflected the nature of these past studies. However, there is a need for an extensive review of studies on the effects of biochar on GHG emissions under field conditions, which provide a more realistic representation of crop production environment than studies conducted under laboratory, greenhouse or mesocosm conditions. Such a review can contribute to improving farmers' confidence in the technology and its adoption. It should be noted that the review by Verhoeven et al. (2017) was based on field-level studies, but it focused solely on N_2O emission. Therefore, we conducted a review of field studies on the effects of soil application of biochar in agriculture on GHG emissions (CO_2, N_2O, and CH_4). The main objective of this review was to synthesize advances on the effect of soil application of biochar on GHG mitigation and identify the research needed to harness this potential toward designing negative emission agricultural systems.

2 | REVIEW APPROACH

Peer-reviewed journal articles published in English before 2022 were searched using the Scopus database (www.scopus.

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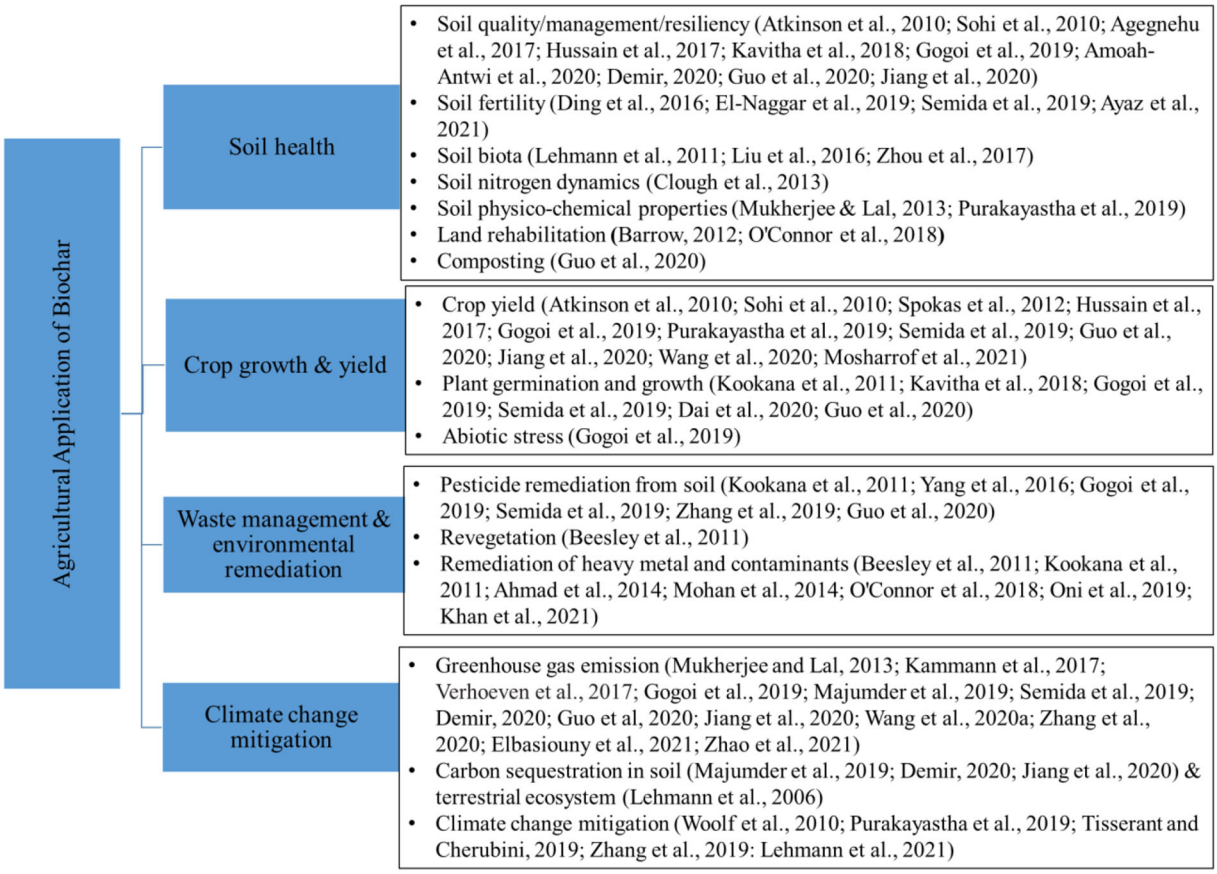


FIGURE 4 Published review papers (in English) on the biochar application to soils of agroecosystem and its impact.

com). The keywords were "biochar" and "soil" and "greenhouse gas emission" or "negative emissions". Only "field" studies (research conducted in field) related to "cropland" were considered. A total of 33 field studies were identified for N_2O (27 from Asia, 5 from North America and 1 from Europe), 31 for CH_4 (26 from Asia and 5 from North America), and 20 for CO_2 (14 from Asia, 5 from North America, and 1 from Africa) (Table 1). When data were reported for multiple years for the same experiment, we reported results as a mean across years rather than by individual year. To make data uniform and comparable, all data on GHG emissions, including extracted from figures, were converted to and presented as Mg C ha^{-1} $year^{-1}$ for CO_2, kg N ha^{-1} $year^{-1}$ for N_2O, and kg C ha^{-1} $year^{-1}$ for CH_4.

3 | ADVANCES IN THE EFFECT OF BIOCHAR APPLICATION TO AGRICULTURAL SOILS ON GREENHOUSE GAS EMISSIONS

Agriculture is a significant contributor to anthropogenic global warming (Lynch et al., 2021). Agriculture, forestry, and other land use contributed an estimated 23% of total anthropogenic GHG emissions between 2007 and 2016 (IPCC, 2019; Laborde et al., 2021). Annual GHG fluxes from land-use and land-use change activities between 2000 and 2010 were estimated at 4.3–5.5 Gt CO_2 eq $year^{-1}$ (Smith et al., 2014). However, integration of biochar technology in land management by farmers has shown potential to significantly reduce such fluxes of GHG (He et al., 2017; Liu, Mao, et al., 2019; Shakoor, Shahzad, et al., 2021). Therefore, biochar application to agricultural soils has grown in appreciation as a strategy to offset GHG emissions and mitigate climate change (Lehmann et al., 2021; Shakoor, Arif, et al., 2021; Sri Shalini et al., 2021). Utilization of agricultural and forestry byproducts and waste for pyrolysis to produce biochar as a soil amendment for agricultural field application is vital for long-term sequestration of atmospheric CO_2 into soils (more than 100 years) and reduction of GHG emissions (Ashiq & Biswas, 2021; Gaunt & Cowie, 2012; Jha et al., 2010). Life cycle-based assessments have shown that biochar production does not necessarily cause significant GHG emissions (Roberts et al., 2010). Instead, field application of biochar can reduce GHG emissions by a maximum of 1.8 Pg CO_2–C equivalent per year (12% of anthropogenic CO_2–C equivalent emissions) (Woolf et al., 2010). Recently, Lehmann et al. (2021) estimated that globally biochar systems could deliver

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TABLE 1 Studies investigating the effect of soil application of biochar on greenhouse gas emissions.

Continent/ country	Soil	Crop	Biochar feedstock	Duration (Year)	Application rate (Soil amendments in Mg ha^{-1}, N-fertilizer in kg ha^{-1})	GHG emissions N_2O Kg–N ha^{-1} year^{-1}	GHG emissions CO_2 Mg–C ha^{-1} year^{-1}	GHG emissions CH_4 Kg–C ha^{-1} year^{-1}	References
Africa									
Nigeria	Sandy clay loam	Rice	Rice husk	2	Control	NA	6.45	NA	Oladele and Adetunji (2021)[A]
					90N	NA	9.98	NA	
					12 biochar	NA	6.81	NA	
					12 biochar + 90N	NA	7.30	NA	
Asia									
China	Clay loam	Tobacco	Tobacco stalks	1	Control	NA	5.74	NA	Tang et al. (2021)
					1 biochar	NA	6.96	NA	
					10 biochar	NA	7.30	NA	
					25 biochar	NA	8.03	NA	
					50 biochar	NA	8.57	NA	
China	NA	Peanut	Cottonseed	1	Control	NA	4.50	−0.52	Tan et al. (2021)
			Rice husk		N	NA	5.75	−0.67	
					9 chicken manure (CM)	NA	6.74	−0.35	
					N + 50 cottonseed biochar	NA	6.16	−0.75	
					N + 50 rice husk biochar	NA	6.28	−0.72	
					CM + 50 CS biochar	NA	7.82	−0.55	
					CM + 50 RH biochar	NA	7.54	−0.61	
China	Silt loam	Rice	Rice straw	1	Control	2.73	59.9	243	Wang, Wang et al. (2019)
					8 steel slag	1.98	54.8	210	
					8 biochar	2.31	49.0	170	
					8 steel slag + 8 biochar	2.17	47.8	161	
China	Luvisol	Tobacco	Peanut shells	1	37.5N	1.34	NA	−2.01	He et al. (2021)
					2.3 biochar + 37.5N	0.94	NA	−2.69	
					2.3 wheat straw-C + 37.5N	1.64	NA	−2.19	
					1.125 biochar + 1.125 straw-C + N	1.46	NA	−2.45	

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TABLE 1 (Continued)

Continent/ country	Soil	Crop	Biochar feedstock	Duration (Year)	Application rate (Soil amendments in Mg ha^{-1}, N-fertilizer in kg ha^{-1})	GHG emissions N_2O Kg–N ha^{-1} year^{-1}	GHG emissions CO_2 Mg–C ha^{-1} year^{-1}	GHG emissions CH_4 Kg–C ha^{-1} year^{-1}	References
China	Lacustrine sediments	Wheat–rice	Rice hulls	2	Control	0.29	NA	49	Wang, Wang et al. (2019)
					270N	0.65	NA	93	
					270N + 25 biochar	0.54	NA	105	
China	Sandy loam	Corn	Corn residue	2	Control	0.946	15.3	−0.89	Yang et al. (2020)
					15 biochar	0.397	11.9	−1.66	
					30 biochar	0.200	12.0	−1.80	
					45 biochar	0.225	10.5	−0.23	
China	Silty clay loam	Rice–wheat	Wheat straw	6	Control	0.18	NA	38.7	Wu, Zhang et al. (2019)
					20 biochar + 0N	0.23	NA	39.9	
					40 biochar + 0N	0.2	NA	37.7	
					0 biochar + 250N	3.53	NA	83.3	
					20 biochar + 250N	2.84	NA	73.9	
					40 biochar + 250N	2.6	NA	68.7	
China	Lacustrine deposits	Rice	Wheat straw	1	Control	18.34	NA	186	Zhang et al. (2010)
					10 biochar + 0N	13.27	NA	306	
					40 biochar + 0N	14.48	NA	311	
					0 biochar + 300N	48.01	NA	206	
					10 biochar + 300N	28.71	NA	199	
					40 biochar + 300N	23.64	NA	318	
China	Clay loam	NA	Corn stalk	1	Control	561	1.91	−22.1	Hou et al. (2020)
					20 biochar	470	1.97	−50.0	
					12 corn straw (CS)	576	2.08	−8.0	
					10 biochar + 6 CS	385	2.01	−47.1	
China	NA	Wheat	Wheat	1	Control	1.00	NA	0.140	Xiang et al. (2015)
					263N	9.95	NA	0.284	

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					225N	7.48	NA	0.271	
					225N + 3.75 biochar	7.41	NA	0.183	
					225N + 7.5 biochar	7.01	NA	0.201	
China	Red soil		Camellia oleifera fruit shell	1	Control	0.92	NA	NA	Deng et al. (2019)
					400N	3.75	NA	NA	
					400N + dicyandiamide	2.12	NA	NA	
					400N + 10 biochar	2.38	NA	NA	
China	Hapli-Udic Cambisol	Corn	Corn stover	2	Control	1.09	4.32	−0.48	Yang et al. (2017)
					2.63 biochar	0.66	5.25	−0.76	
					7.5 corn stover	0.84	9.99	−1.07	
China	NA	Rice	Rice straw	2	210N	NA	20.9	75	Sui et al. (2016)
					210N + 5.1 rice straw	NA	19.0	238	
					210N + 1.8 biochar	NA	21.4	31	
					210N + 14.8 biochar	NA	29.4	66	
					210N + 29.6 biochar	NA	19.9	52	
China	Calcareous	Corn	Wheat straw	1	control	0.76	6.00	−0.43	Zhang, Liu et al. (2012)
	Fluvo-aquic loamy				20 biochar + 0N	0.87	6.18	1.30	
					40 biochar + 0N	0.83	6.72	−1.99	
					0 biochar + 300N	4.41	5.43	−1.41	
					20 biochar + 300N	3.94	5.49	−0.47	
					40 biochar + 300N	2.57	5.29	−0.40	
China	Hydroagric Stagnic Anthrosol and an Entic Halpudept	Rice[C]	Wheat straw	2	0 biochar + 300N	9.87	4.00	148	Zhang, Bian et al. (2012)
					10 biochar + 300N	6.52	3.78	139	
					20 biochar + 300N	5.06	3.74	333	
					40 biochar + 300N	4.53	4.02	219	
China	Paddy soil	Rice–wheat	Wheat straw	1	0 biochar + 300N	11.5	NA	113	Zhang et al. (2013)

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					10 biochar + 300N	11.8	NA	109	
					20 biochar + 300N	7.0	NA	147	
					40 biochar + 300N	6.8	NA	118	
China	Littoral clay salt soil	Rice	Wheat straw	1	Control	0.71	NA	406	Sun et al. (2020)
					20 biochar + 0N	0.87	NA	501	
					40 biochar + 0N	0.93	NA	532	
					0 biochar + 300N	1.64	NA	520	
					20 biochar + 300N	1.79	NA	560	
					40 biochar + 300N	2.29	NA	530	
China	Silty clay loam	Corn	Tobacco straw	2	0 wheat straw + 225N[D]	6.70	NA	NA	Li et al. (2021)
			Residues		6 straw + 225N	10.60	NA	NA	
					8 biochar + 6 straw + 225N	7.07	NA	NA	
					0 straw + 158N	3.90	NA	NA	
					4.8 straw + 158N	5.16	NA	NA	
					8 biochar + 4.8 straw + 158N	3.58	NA	NA	
China	Loess deposits, silt clay loam	Wheat	Wheat straw	2	Control	0.64	9.9	−3.0	Zhang et al. (2017)
					375N	2.80	10.3	−2.4	
					8 wheat straw + 375N	2.70	11.8	−2.2	
					8 biochar + 375N	1.70	10.1	−1.1	
					16 biochar + 375N	1.50	9.8	−4.4	
China	Sandy loam	Rice	Corn straw	3	Control	14.3	NA	132	Liu, Liu et al. (2016)
					6 corn straw	14.6	NA	1045	
					2.4 biochar	12.8	NA	99	
China	Clay loam lacustrine sediment	Rice–wheat	Wheat straw	2	Control	1.21	NA	348	Ma, Li et al. (2019)
					0 biochar + 330N	2.45	NA	419	
					20 biochar + 330N	2.71	NA	389	
China	Sandy loam texture	Barley-cotton	Wheat straw	2	Control	42.2	24.2	3.1	Ma et al. (2021)

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TABLE 1 (Continued)

Continent/ country	Soil	Crop	Biochar feedstock	Duration (Year)	Application rate (Soil amendments in Mg ha^{-1}, N-fertilizer in kg ha^{-1})	GHG emissions N_2O Kg–N ha^{-1} year^{-1}	GHG emissions CO_2 Mg–C ha^{-1} year^{-1}	GHG emissions CH_4 Kg–C ha^{-1} year^{-1}	References
					3.5 straw biochar	26.2	21.8	1.5	
					5 straw + inoculant	30.2	32.6	10.1	
					5 straw	32.7	29.7	7.6	
					5 straw mulching	38.2	26.0	4.9	
China	Clay loam	Rice	Wheat straw	3[E]	Control	NA	NA	73	He et al. (2020)
					300N	NA	NA	113	
					7.5 biochar	NA	NA	65	
					7.5 biochar + 300N	NA	NA	85	
China	Alkaline sandy loam	Corn–corn	Rice straw	5	Control	1.36	NA	NA	Liu et al. (2020)[F]
					2.5 biochar	0.97	NA	NA	
					6.75 biochar	0.91	NA	NA	
					11.3 biochar	0.84	NA	NA	
China	Regosol	Vegetables[G]	Rape straw	2	Control	2.81	36.8	1.1	Qi et al. (2020)
					300N	7.84	34.2	0.3	
					[H]Biochar + 300 N	6.14	47.9	−0.5	
		Rice	Rape straw	2	Control	0.24	20.9	142	
					150N	0.87	37.9	28.7	
					10 biochar + 106N	1.72	32.8	105	
China	Slightly acidic loam	Rice	Wheat straw	5	Control	4.87	NA	95	Liu, Zhou et al. (2019)[I]
					20 biochar + 0N	1.52	NA	93	
					40 biochar + 0N	1.38	NA	80	
					0 biochar + 150N	12	NA	89	
					20 biochar + 150N	9.23	NA	72	
					40 biochar + 150N	5.37	NA	61	
China	Ferralsol	Chestnut	Bamboo leaf	2	Control	2.60	15.75	2.80	Lu et al. (2019)
					2 biochar	2.35	16	2.95	
					5 biochar	2.20	16.25	3.45	
					10 biochar	2.05	16.55	3.60	
Japan	Clay loam	Broccoli	Palm shell	1	Control	5.91	4.92	1.31	Basalirwa et al. (2020)

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TABLE 1 (Continued)

Continent/ country	Soil	Crop	Biochar feedstock	Duration (Year)	Application rate (Soil amendments in Mg ha^{-1}, N-fertilizer in kg ha^{-1})	GHG emissions N_2O Kg–N ha^{-1} year^{-1}	GHG emissions CO_2 Mg–C ha^{-1} year^{-1}	GHG emissions CH_4 Kg–C ha^{-1} year^{-1}	References
					10 biochar	7.39	4.54	1.32	
					20 biochar	4.91	4.88	1.42	
					40 biochar	3.27	4.40	1.58	
Korea	Silt loam	Rice	Sesame straw	1	Control	3.9	NA	383	Kang et al. (2016)
					90N	15.2	NA	370	
					1 biochar + 90N	9.3	NA	229	
Korea	Sandy loam	Rice	Barley straw	1	Control	4.9	NA	1278	Kang et al. (2021)
					90N	28.0	NA	1247	
					20 biochar	7.6	NA	767	
					20 barley straw (BS)	11.3	NA	1180	
					10 biochar + 10 BS	9.7	NA	782	
Pakistan	Silt loam	Corn–wheat	Acacia tree prunings	1	Control	0.10	NA	NA	Dawar et al. (2021)[J]
					150N	1.35	NA	NA	
					5 biochar	0.42	NA	NA	
					10 biochar	0.44	NA	NA	
					5 biochar + 150N	1.01	NA	NA	
					10 biochar + 150N	0.93	NA	NA	
Europe									
Denmark	Sandy loam	Oilseed rape (*Brassica napus* L.)	Wheat straw and pig manure	2	Control	2.20	NA	NA	Thers et al. (2020)[K]
					15 manure biochar	2.32	NA	NA	
					15 straw biochar	2.06	NA	NA	
North America									
Canada	Loamy sand	Corn	Yellow pine wood	2[L]	Dairy manure (DM)[M]	4.53	18.5	16.0	Ashiq et al. (2020)
					DM + 20 biochar	0.22	15.4	−9.8	
					Inorganic N (IN)[N]	4.06	15.9	13.4	
					IN + 20 biochar	0.53	11.6	−25.7	
					0N	2.30	12.4	−0.5	
USA	Sandy soil	Grassland	Pinewood	3	Control	7	46	3	Lu et al. (2020)

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TABLE 1 (Continued)

Continent/ country	Soil	Crop	Biochar feedstock	Duration (Year)	Application rate (Soil amendments in Mg ha^{-1}, N-fertilizer in kg ha^{-1})	GHG emissions N_2O Kg–N ha^{-1} year^{-1}	GHG emissions CO_2 Mg–C ha^{-1} year^{-1}	GHG emissions CH_4 Kg–C ha^{-1} year^{-1}	References
					20 biochar	7	42	1	
					240 biosolids-N(BN)	9	48	4	
					240 BN + 20 biochar	8	43	0.5	
					160N	5	44	2	
					160N + 20 biochar	5	40	1	
	USA	Loess-derived Mollisols	Corn	Mixture of hardwood and soft wood	2	Control 9.3 biochar	NA NA	10.4 10.2	2.15 1.92	Fidel et al. (2019)[O]
	USA	Sandy loam	Corn−soybean −soybean	Pine, switch grass, and corn stover	3	Control	5.2	9.6	−1.3	Abagandura et al. (2019)
						10 corn stover biochar	3.5	9.2	−1.3	
						10 pine biochar	3.3	9.0	−1.6	
						10 switch grass biochar	3.3	9.0	−1.0	
						10 dairy manure	3.4	8.2	−1.4	
						5 dairy manure + 5 pine biochar	2.7	7.9	−1.2	
		Clay loam	Corn−soybean −soybean	Pine, switch grass, and corn stover	3	Control	8.1	8.7	0.4	
						10 corn stover biochar	6.8	8.8	0.7	
						10 pine biochar	6.6	9.0	0.0	
						10 switch grass biochar	6.9	9.6	1.2	
						10 dairy manure	4.4	6.5	1.9	
						5 dairy manure + 5 pine biochar	4.0	6.1	1.7	
	USA	Silty clay loam	Corn	Hardwood sawdust	2	Control	8.8	10.4	NA	Sistani et al. (2019)
						21 biochar	9.6	9.9	NA	
						224 fertilizer	18.9	9.2	NA	
						Poultry litter (eq to 224 kg N)	25.3	12.0	NA	
						Biochar + fertilizer	14.6	8.2	NA	
						Biochar + poultry litter	15.3	10.6	NA	
	USA	Sandy loam	Ryegrass (*Lolium multiflorum* L.)	Pine	1	Manure (410 kg N)	19	NA	8.8	Angst et al. (2014)

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TABLE 1 (Continued)

Continent/ country	Soil	Crop	Biochar feedstock	Duration (Year)	Application rate (Soil amendments in Mg ha^{-1}, N-fertilizer in kg ha^{-1})	GHG emissions N_2O Kg–N ha^{-1} year^{-1}	GHG emissions CO_2 Mg–C ha^{-1} year^{-1}	GHG emissions CH_4 Kg–C ha^{-1} year^{-1}	References
					Manure (410 kg N) + 5.7 biochar	20	NA	9.8	
					Manure (410 kg N) + 18.8 biochar	16	NA	5.4	

Abbreviation: NA, not available.

[A]Average of 2 years estimated from a figure.

[B]Total gas emissions for the fallow and crop seasons broccoli residue + 0, 10, 20, and 40 Mg ha^{-1} biochar.

[C]Average of rice and wheat seasons.

[D]Fertilizers were spread by hand as basal dressing after straw residues were chopped then were mixed with straw into the soil to a depth of 15 cm. An amount of 6 and 2 Mg ha^{-1} biochar were applied to corn and wheat, respectively, in 2018.

[E]Average of 2 years data.

[F]The values are average of 5 years cumulative N_2O emissions. A total of 10 crops grown in 5 years (5 crops of wheat and 5 crops of corn).

[G]Data estimated from figure. Methane data were for the fifth year, and nitrous oxide data were for the second year of biochar application, when peaks were observed.

[H]10 biochar + 256N (chili), 300N (lettuce), and 0N (cabbage).

[I]GHG emission values estimated from figure.

[J]Total for 6 months.

[K]224 kg N ha^{-1} added to all treatments.

[L]Cumulative GHG emission per season for average (high and low) manure application over 2 years.

[M]Dairy manure 30,000 L ha^{-1}.

[N]Nitrogen 115 kg ha^{-1}.

[O]Values estimated from figure.

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emission reductions of 3.4–6.3 Pg CO_2 equivalent. Mechanisms by which biochar can reduce GHG from soils include increased soil aeration (Karhu et al., 2011; Wang et al., 2021), moderation of soil temperature (Feng et al., 2021) and soil pH (Berihun et al., 2017), and evolution of organic compounds (Zhang, Bian, et al., 2012) that constrain the activity of denitrifying bacteria and favor the activity of CH_4-oxidizing bacteria. The nature and directions of the interactions among biochar properties, soil type, and biochemical processes (respiration, nitrification) may also contribute to the variable effects of biochar on GHG emissions (Chang et al., 2016).

3.1 | Carbon dioxide emissions

In agroecosystems, CO_2 is released mainly from soil respiration or burning of plant residue (Bairwa et al., 2020; Janzen, 2004). Several studies have reported that the release of CO_2 from soils can be reduced with biochar amendment (Ashiq et al., 2020; Schmidt et al., 2021; Yang et al., 2020). Across field studies, biochar increased carbon dioxide emissions by 1.9% (−31% to 49%) compared to without biochar. Out of 29 field studies that evaluated the effects of biochar alone on CO_2 emissions, 14 reported a reduction in CO_2 emissions (range: −1% to −31%, mean: −11%) when biochar was applied compared to without biochar (Table 1). Carbon dioxide emission reductions were not proportional to biochar application rates (Figure 5).

Reduction in CO_2 emissions after biochar application is usually observed in soils with lower SOC content (e.g., 21 to 24 g SOC kg^{-1} soil in the 0–10 cm depth; Kimetu & Lehmann, 2010). Here, biochar application reduced the rate of CO_2 emission by −27%. In the same study, CO_2 emission was greater in soils with higher SOC content (i.e., 58 g SOC kg^{-1} in the 0–10 cm depth). In a sandy loam soil under a dry climate regime, CO_2 emitted during the corn (*Zea mays* L.) growing season decreased at biochar application rate of 15 Mg ha^{-1} (Yang et al., 2020). Total CO_2 emission for two growing seasons averaged 5.1, 4.0, 4.0, and 3.6 Mg C ha^{-1} in plots treated with biochar at the rates of 0, 15, 30, and 45 Mg ha^{-1}, respectively. Biochar application to corn field was found to generally accelerate soil respiration early in the growing season but restricted it later in the growing season (Qi et al., 2020; Yang et al., 2020). Water soluble surface-adsorbed labile C released from newly applied biochar may have undergone microbial decomposition following biochar application and may have promoted CO_2 emissions during the early period, until 6 days after application (Smith et al., 2010). In a study involving biochar application in consecutive years, it was found that biochar application reduced CO_2 emissions by −18 to −25% during the first growing season and by −19% to −41% during the second growing season compared to soils without biochar (Yang et al., 2020). This suggests that the biochar applied in the first year may have a carryover effect on limiting CO_2 emissions in the following years. Fifteen field studies (out of 29) reported that biochar amendments had either no effects or increased soil CO_2 emissions (Figure 5). The increase in CO_2 emission may be due to the mineralization of the labile C fractions present in the biochar (Shen et al., 2017; Zimmerm et al., 2011), increase in soil temperature leading enhanced decomposition of SOM (Lehmann et al., 2011), and balanced soil C/N ratio resulting improved soil microbial C mineralization (Jianwen et al., 2004) and enhanced priming effect on soil C mineralization (Zimmerman et al., 2011).

Out of 18 field observations that evaluated the effects of biochar when applied in combination with N-fertilizer, 11 reported a reduction in CO_2 emissions by an average of −10% (range: −2 to −27%) when biochar was applied in combination with N-fertilizer compared to N-fertilizer alone (Table 1, Figure 6a). Similarly, nine field observations that reviewed the effects of biochar when applied in combination with soil amendments (e.g., manure, biosolids, steel slug, or straw), and seven reported a reduction in CO_2 emissions by an average of −9% (range: −4% to −17%) when biochar was applied in combination with amendments compared to amendments alone (Table 1, Figure 7a). About 48% of the field observations reviewed reported reduction in CO_2 emission when biochar was applied compared to without biochar. When biochar was applied in combination with either fertilizer or other soil amendments, reduction in CO_2 emission increased to 61% or 78%, respectively, compared to biochar alone. This indicates that biochar is more effective in reducing CO_2 emissions when combined with either fertilizer or other soil amendments.

3.2 | Methane emissions

Methane is produced when organic materials decompose under oxygen-deprived conditions (Mosier et al., 1998). Biochar application to soils has the potential to reduce CH_4 emissions (Sriphirom et al., 2020, 2021; Yang et al., 2020; Zhao et al., 2021), especially from flooded or acidic soils (Wang, Shen, et al., 2019; Wu, Song, et al., 2019). The efficiency of biochar in mitigating CH_4 depends on soil moisture (Cong et al., 2018), soil texture (Cong et al., 2018), soil aeration (Nan et al., 2021), soil pH (Gross et al., 2021; Nan et al., 2021), soil organic C content (Cong et al., 2018; Kimetu & Lehmann, 2010), and biochar feedstock (Abagandura et al., 2019) and properties (Jeffery et al., 2016). Biochar affects CH_4 emission by regulating functional microorganisms (especially methanotroph communities) and soil physicochemical properties (such as aggregation, porosity, moisture, soil temperature, pH, and N) affecting CH_4 transport through soils and oxidation by methanotrophs (He et al., 2021; Zhao et al., 2021). Other enhancements of soil physical

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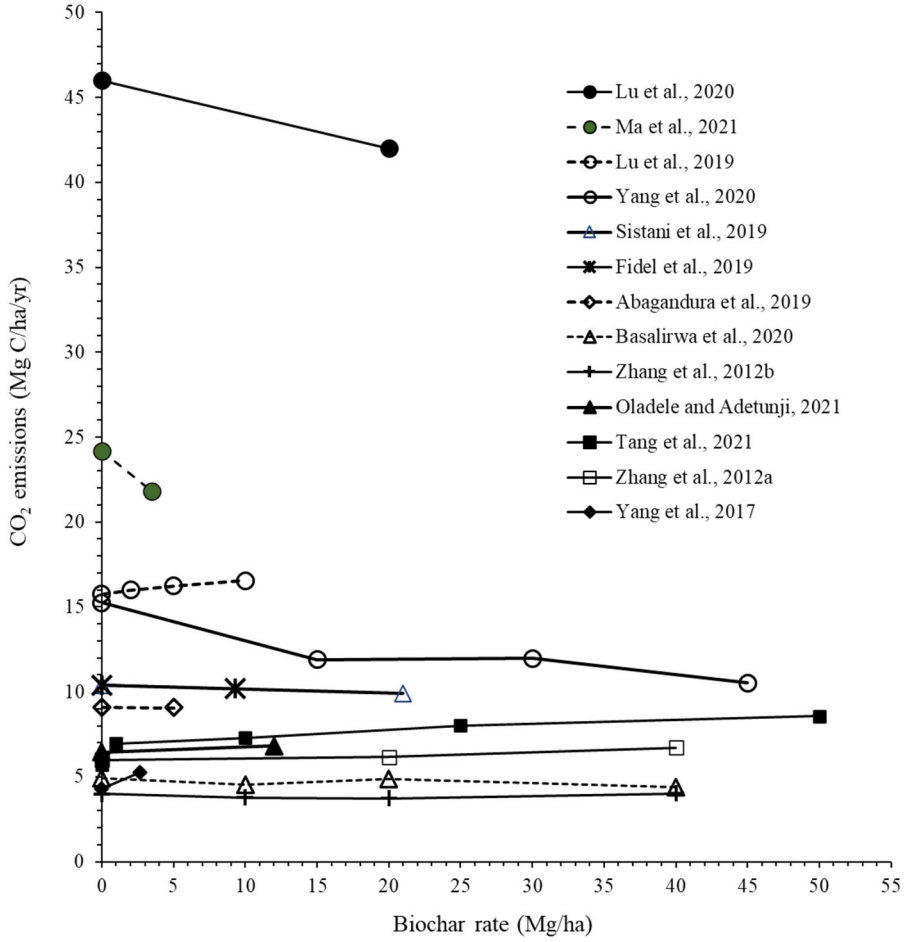


FIGURE 5 Effect of soil application of biochar on CO_2 emissions.

properties, such as bulk density and porosity, in response to biochar application also contribute to increase in soil permeability, favoring oxidation of CH_4 (Liu & Greaver, 2009; Mer & Roger, 2001; Oladele, 2019).

Across field studies, biochar decreased CH_4 emissions by −3% (−358% to 400%) compared to without biochar. Out of 33 field studies that evaluated the effects of soil application of biochar alone on CH_4 emissions, 15 reported average reduction in CH_4 emissions by −62% (range: −2% to −358%) when biochar was amended to soil compared to without biochar (Table 1). Reduction in CH_4 emissions with soil application of biochar was evident mostly at locations exhibiting low emission rates (Figure 8a). However, increased CH_4 emissions with biochar application compared to zero biochar application were evident for cases with much higher emission rates (Figure 8b). All the high CH_4 emissions (8B) were from rice cultivation, where soil is mostly flooded and under anaerobic conditions, and most of the low CH_4 emissions were from upland fields where soils are under aerobic conditions (8A), usually resulting in a net CH_4 sink. Additionally, increasing CH_4 emissions with biochar compared to without biochar application especially for cases with much higher emission rates may be due to (1) labile components of biochar that can be decomposed and become the predominant substrates for methanogens, thus increasing CH_4 emissions (Knoblauch et al., 2011), and (2) increase in soil pH after biochar amendment in rice paddies resulting in enhanced methanogenic activity (Troy et al., 2013; Yu et al., 2013).

The effectiveness of biochar in mitigating CH_4 emissions was more common in field studies when biochar was applied in combination with chemical fertilizer or soil amendments (e.g., manure, biosolid, straw, and steel slag). Out of 33 field observations that evaluated the effects of biochar when applied in combination with chemical fertilizer, 21 reported a reduction in CH_4 emissions by an average of −54% (range: −3% to −292%) when biochar was applied in combination with N-fertilizer compared to N-fertilizer alone (Table 1, Figure 6b). Out of 12 field studies that evaluated the effects of biochar when applied in combination with soil amendments, 11 reported a reduction in CH_4 emissions by an average value of −91% (range: −10% to −488%) when biochar was applied in combination with soil amendments compared to soil amendments alone (Table 1, Figure 7b). The CH_4 emission rates did not

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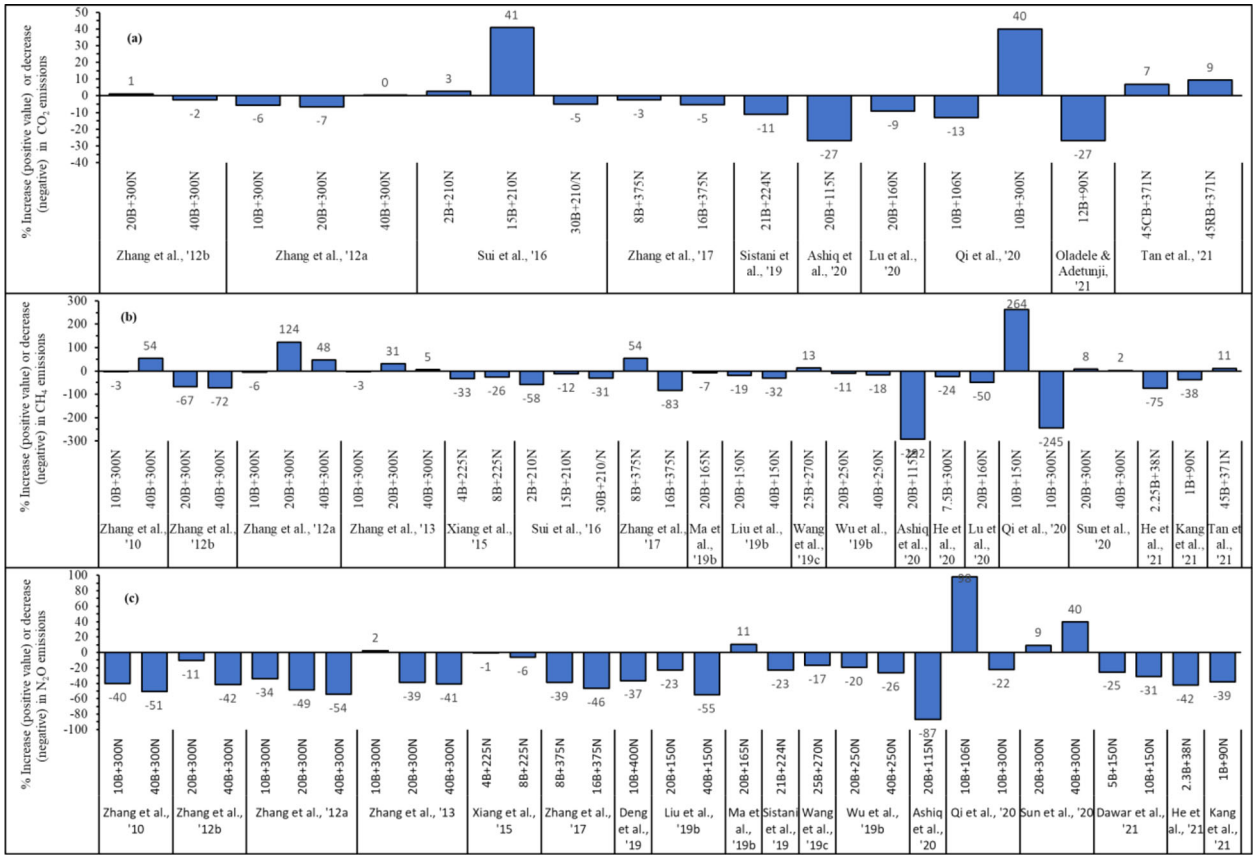


FIGURE 6 Changes (%) in (a) CO_2 (b) CH_4 and (c) N_2O emissions from combination of fertilizer plus biochar addition compared to fertilizer alone. The values above and below the bars are percent CO_2, CH_4, and N_2O emission. Biochar treatments are listed above each reference. B, biochar (Mg ha^{-1}); CB, cotton seed biochar (Mg ha^{-1}); *N*,*N*-fertilizer (Mg ha^{-1}); RB, rice husk biochar (Mg ha^{-1}).

consistently decrease with increasing rate of biochar application (Figure 8).

Two mechanisms, decrease in CH_4 emission and increase in CH_4 oxidation by methanotrophs, may be operational after biochar application (Karhu et al., 2011). Biochar application may have improved the growth and abundance of CH_4-oxidizing bacteria (methanotrophs), likely resulting in CH_4 oxidation dominating CH_4 production by methanogens (Feng et al., 2012; Wang, Shen, et al., 2019). Decrease in CH_4 emission may also be due to increase in soil porosity and aeration after field application of biochar, resulting in more aerated soil and increased oxidation of CH_4 (Van Zwieten et al., 2009). Improvement in soil aeration (Van Zwieten et al., 2009) and high surface area of biochar (Leng et al., 2021) may contribute to reduction in CH_4 emissions.

In a silty clay loam soil, CH_4 emissions decreased up to −11% when N-fertilizer (250 kg ha^{-1}) was applied in combination with biochar (20–40 Mg ha^{-1}) compared to N-fertilizer alone (Wu, Song, et al., 2019). In a clay loam soil, even greater reduction (−24%) in CH_4 emissions was recorded when biochar (7.5 Mg ha^{-1}) was applied in combination with N-fertilizer (300 kg ha^{-1}) compared to N-fertilizer alone (He et al., 2020). In another study on a clay loam soil, CH_4 emissions were decreased by −17% in the first season and −23% in the second season when N-fertilizer (100 kg N ha^{-1}) was applied in combination with one-time application of biochar (10 Mg ha^{-1}) (Sriphirom et al., 2021). As noted with CO_2 emission, this suggests that biochar applied in the first year can continue to reduced CH_4 emission in the succeeding year. Methane emissions from manure application were significantly reduced by an average of −91% (range: −10% to −488%) when applied in combination with biochar compared to manure application alone (Figure 7b). In a corn production system, the combination of pine wood (*Pinus spp.*) biochar with either dairy manure or N-fertilizer resulted in significant reduction in CH_4 emission (−200% and −172%, respectively) compared to either manure or fertilizer applied alone (Ashiq et al., 2020). Similarly, CH_4 emission following soil amendment with chicken manure (9 Mg ha^{-1}) was decreased by −67% when combined with biochar (50 Mg ha^{-1}) compared to emissions recorded when chicken manure was applied alone (Tan et al., 2021). Overall, reduction in CH_4 emission was observed for different soil types, N-fertilizer formulations (conventional and slow release; Dong et al., 2021), and irrespective of biochar application rate and feedstock [cotton seed husk (*Gossypium* spp.), rice husk (*Oryza sativa* L.), Tan et al.,

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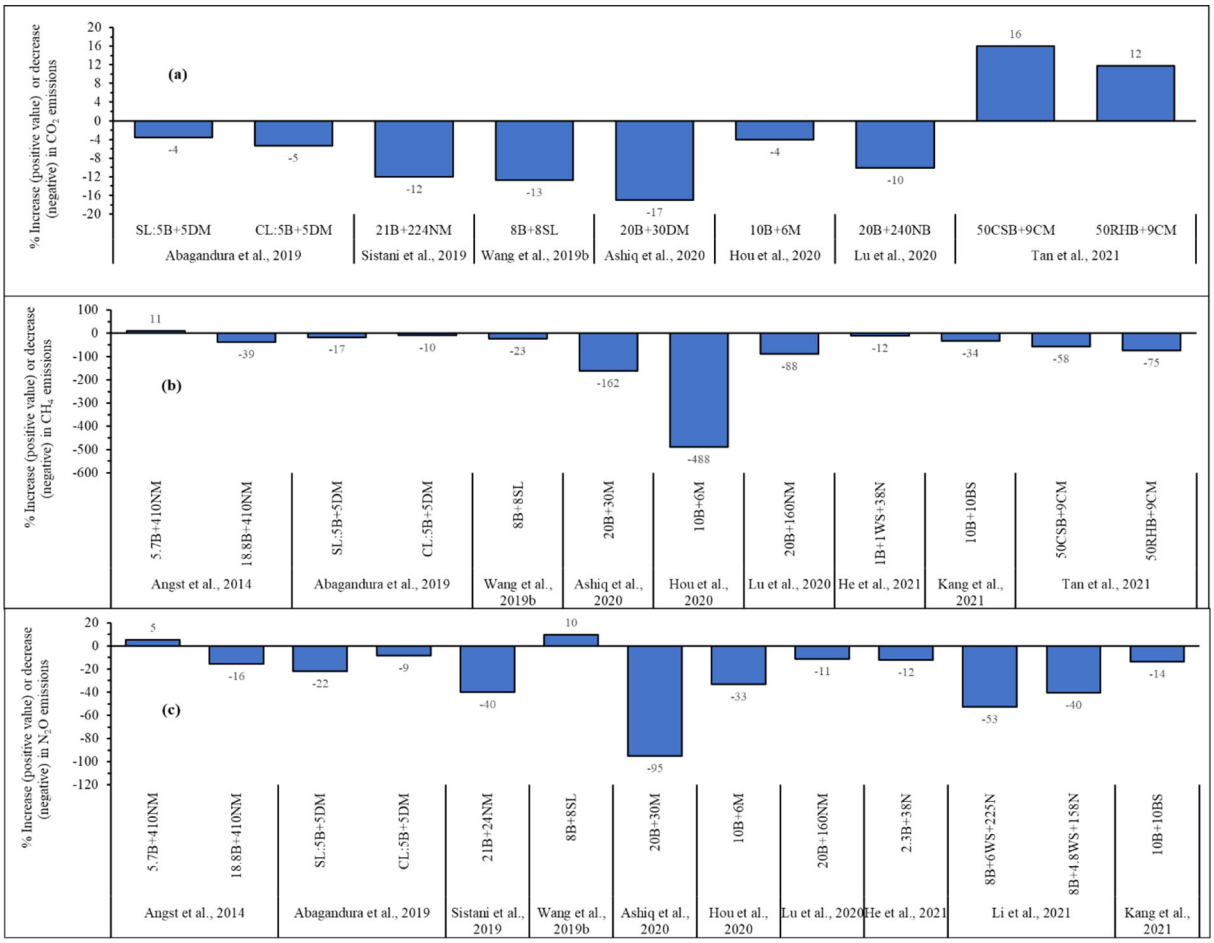


FIGURE 7 Changes (%) in (a) CO_2, (b) CH_4, and (c) N_2O emissions from combination of biochar plus manure compared to manure alone. *Note*: The values above and below the bars are percent CO_2, CH_4, and N_2O emission values. Biochar treatments are listed above each reference. B, biochar (Mg ha^{-1}); B-C, biochar-carbon (Mg ha^{-1}); B-S, barley straw (Mg ha^{-1}); CL, clay loam; CM, chicken manure (Mg ha^{-1}); CS, corn straw (Mg ha^{-1}); CSB, cotton seed biochar (Mg ha^{-1}); DM, dairy manure; N, N-fertilizer (Kg ha^{-1}); NB, N from biosolid (Kg ha^{-1}); NM, nitrogen from manure (Kg ha^{-1}); RHB, rice husk biochar (Mg ha^{-1}); SL, silt loam; SS, steel slag (Mg ha^{-1}); WS, wheat straw (Mg ha^{-1}); WS-C, wheat straw-carbon (Mg ha^{-1}).

2021; pine, Ashiq et al., 2020; mangrove (*Rhizophora apiculata* L.), Sriphirom et al., 2021]. These results likely stem from a biochar-induced improvement in soil aeration (Chen et al., 2017; Sonoki et al., 2013).

Straw-derived biochar applied as a soil amendment at the rate of 24 and 48 Mg ha^{-1} in double rice cropping system (two rice crops in a year) consistently decreased annual CH_4 emissions by −20% to −51% in a 4-year field experiment (Wang, Wang, et al., 2019). Similar reduction in CH_4 emissions from paddy soil was observed by Shen et al. (2014) with straw-derived biochar applied at the rate of 48 Mg ha^{-1}. In another study, corn residue-derived biochar applied at the rate of 15 and 30 Mg ha^{-1} in a sandy loam soil reduced CH_4 emission by −124% and −132%, respectively, compared to without biochar (Yang et al., 2020). Application of birch-derived biochar (9 Mg ha^{-1}) to a wheat (*Triticum aestivum* L) field, on well-drained silt loam soil, decreased CH_4 emissions (increased CH_4 uptake) by −96% (Karhu et al., 2011). Freeze–thaw cycles can promote the decomposition of soil organic matter especially in cold region encouraging GHG emissions (Risk et al., 2013). However, application of 20 Mg ha^{-1} of biochar in this environment was found to reduce CH_4 emissions during the soil thawing period (Hou et al., 2020). In a rice production system on clay loam soil, Nan et al. (2020) observed that annual application of 2.8 Mg ha^{-1} of biochar (for four consecutive years) reduced CH_4 emission by −41% as compared to a reduction of −38% with a single large application of 22.5 Mg ha^{-1}. Small but repeated applications of biochar have been found to suppress CH_4 emissions more efficiently than a single large application (Nan et al., 2020). The authors concluded that split application could be a strategy to improve biochar efficiency, increase its benefit-to-cost ratio, and thus its financial attractiveness.

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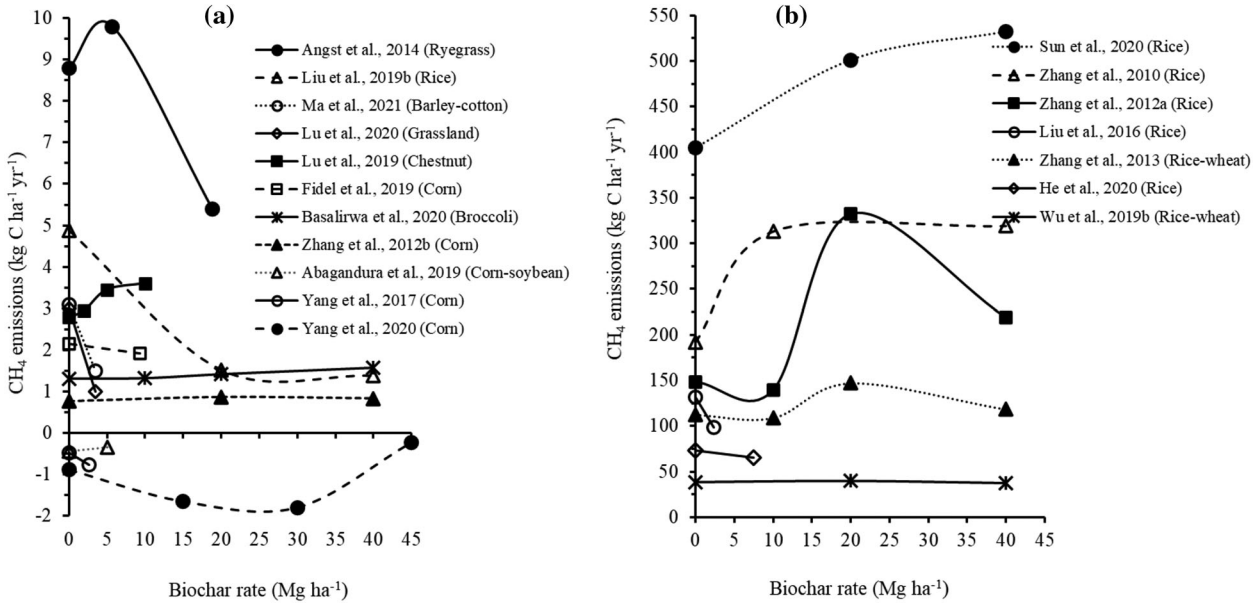


FIGURE 8 Effect of field application of biochar on methane (CH_4) emissions compared to no biochar: (a) studies with emissions <25 kg C ha^{-1} $year^{-1}$ and (b) studies with emissions >25 kg C ha^{-1} $year^{-1}$.

However, studies have also reported increased, or no effect of biochar applied as soil amendment on CH_4 emission (Table 1). The increase in CH_4 emissions may be due to improved methanogenic activities as a result of increased soil moisture levels (Yu et al., 2013) and increased availability of biochar-associated labile organic compounds to methanogens, thus promoting CH_4 production (Knoblauch et al., 2011). No effect of biochar may be because the study was conducted on an upland field that experienced mostly aerobic conditions and the addition of biochar further increased soil aeration (Basalirwa et al., 2020; Karhu et al., 2011). Studies have shown that high rates (20–45 Mg ha^{-1}) of biochar application to rice field resulted in increased CH_4 emissions, whereas net CH_4 uptake by soil was measured with a lower rate (10 Mg ha^{-1}) of biochar application (Ma, Li Liu, et al., 2019; Yang et al., 2020; Zhang et al., 2019). High rate, 45 Mg ha^{-1} biochar, was reported to increase CH_4 emissions in the first year but reduced emissions during the second year, possibly due to progressive degradation of the added biochar (Yang et al., 2020).

3.3 | Nitrous oxide emissions

Biochar application has been suggested as a strategy to reduce N_2O emissions from agricultural soils while increasing SOC stocks (Cayuela et al., 2014; Rittl et al., 2021; Zhang et al., 2019, 2020). Across field studies, biochar decreased N_2O emissions by −18% (range: −79% to 56%) compared to without biochar. Out of 37 field studies that have evaluated the effects of biochar alone on N_2O emissions, 27 reported an average N_2O emission reduction of −33% (range: −6% to −79%) compared to fields without biochar (Table 1). In contrast to the trends observed with CO_2 and CH_4, N_2O emission showed a trend of decreasing emission with increasing rate of biochar application (Figure 9). Studies have reported reductions in N_2O emissions of up to −58% with field application of 10 to 40 Mg ha^{-1} of biochar (Cayuela et al., 2015; Verhoeven et al., 2017; Zhang et al., 2020). Reductions in N_2O emissions following biochar application have been reported for the first 2 years in rice fields established on slightly acidic loamy soil (Liu, Zhou, et al., 2019) and in wheat–corn cropping system on alkaline sandy loam soils (Liu et al., 2020). In a corn production system, the decrease in N_2O emissions was −71% to −110% in the first and −39% to −47% in the second year of biochar application (Yang et al., 2020). Kalu et al. (2021), in a nutrient deficient Umbrisol with a loamy sand texture, reported that the effect of biochar on N_2O emissions reduction (an average of −30% reduction compared to no biochar) persisted for at least 7 years after the application. However, long-term effects of biochar on N_2O emissions need to be further validated on-farm in different soil types and environmental conditions. Even a relatively low rate of biochar (10 Mg ha^{-1}) was shown to reduce N_2O emission by −36% in oilseed camellia (*Camellia oleifera* L.) fields (Deng et al., 2019). Similar rates and trend in N_2O emission reduction were observed by Agegnehu et al. (2016) and Zhang, Bian et al. (2012). Several mechanisms have been reported to explain the N_2O emission reductions in biochar-amended soils including: (1) increased adsorption of NO_3 (Singh et al., 2010; Van Zwieten et al., 2010), (2) improved soil physiochemical properties such as better soil aeration, constraining

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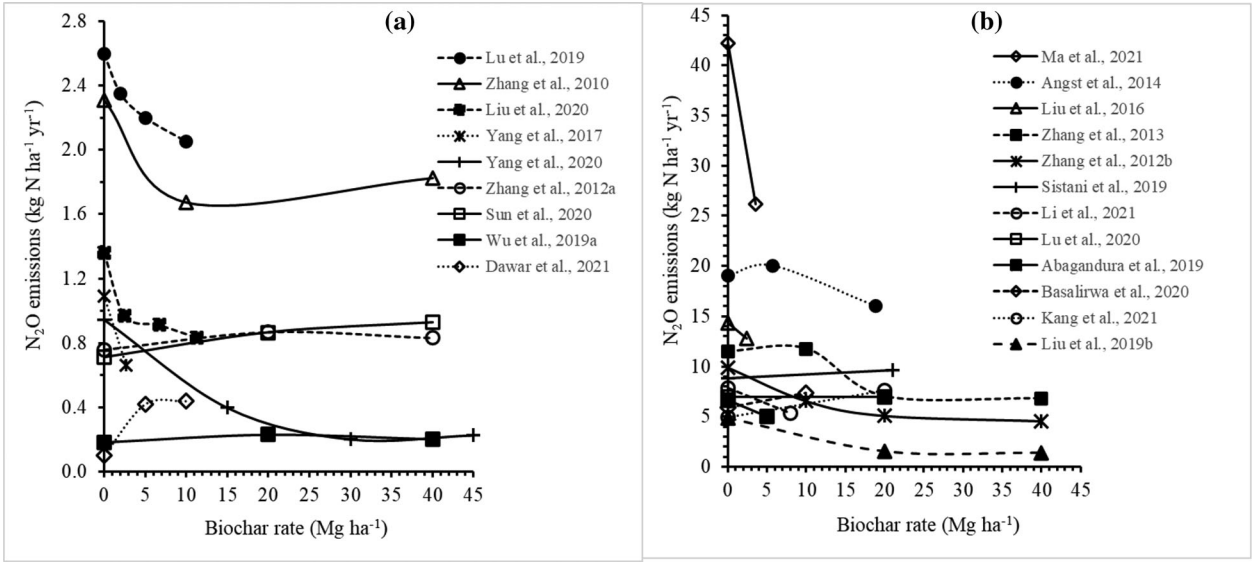


FIGURE 9 Effect of field application of biochar on nitrous oxide (N_2O) emissions compared to no biochar: (a) studies with emissions <4 kg C ha^{-1} $year^{-1}$ and (b) studies with emissions >4 kg N_20–N ha^{-1} $year^{-1}$.

the activity of denitrifiers (Arif et al., 2020; van Zwieten et al., 2010), (3) decreased organic substrates availability to soil microorganisms (He et al., 2021; Kraus et al., 2018; Zhang et al., 2021), resulting in lower activity by denitrifying (Wang, Yang, et al., 2020) and ammonia oxidizing bacteria (Liu et al., 2014), (4) increased soil pH, making conditions more suitable for microbial respiration (van Zwieten et al., 2014) and N_2O reductase (Van Zwieten et al., 2010), thus a lower amount of N_2O in the N gas products, (5) inhibition of N transformation processes at the nitrification stage (Zhang, Bian, et al., 2012, Zhang, Liu, et al., 2012; Das et al., 2014), (6) enhanced the immobilization of mineral N by soil microbes due to the high C:N ratio of biochar, (7) increased adsorption of N_2O on biochar due to large surface area of biochar (He et al., 2019; Lentz et al., 2014), and (8) presence of inhibitory substances such as ethylene that suppress soil microbial activity (Spokas et al., 2010). Nitrogen fertilizers and organic amendments are known to stimulate N_2O emissions from soils (Aliyu et al., 2021; Charles et al., 2017). Nitrous oxide emission rates, however, have been found to significantly decrease when these N sources are applied in combination with biochar (Figures 6c and 7c). Out of 31 field studies reviewed and that examined the effect of biochar application in combination with fertilizer on N_2O emissions, 26 reported an average N_2O emissions reduction of −35% (range: −1% to −87%) (Table 1, Figure 6c). In tobacco (*Nicotiana tabacum* L.) fields, N_2O emissions were found to decrease by −30% when N-fertilizer (38 kg N ha^{-1}) was applied in combination with peanut (*Arachis hypogaea* L.) shells biochar (2.25 Mg ha^{-1}) than N-fertilizer alone (He et al., 2021). In rice fields, application of rice straw biochar (40 Mg ha^{-1}) in combination with N-fertilizer (180 kg N ha^{-1}) reduced N_2O emissions by −62% compared to N-fertilizer alone (Aamer et al., 2021). Similarly, in a rice–wheat rotation system, combined application of wheat straw-derived biochar (20 or 40 Mg ha^{-1}) and N-fertilizer (250 kg N ha^{-1}) significantly reduced N_2O emissions by up to −26% compared to N-fertilizer alone (Wu, Song, et al., 2019).

Out of 13 studies that examined the effect of biochar application in combination with soil amendments other than N-fertilizer on N_2O emissions, 11 reported an average N_2O emissions reduction of −31% (range: −9% to −95%) compared to soil amendment alone (Table 1, Figure 7c). Dairy manure or inorganic N-fertilizer when applied to a corn production system in combination with yellow pine wood biochar, significantly reduced N_2O emissions by −90% and −86%, respectively, compared to manure or fertilizer alone (Ashiq et al., 2020; Table 1). In another study, corn stalk (6 Mg ha^{-1}) when applied in combination with biochar (10 Mg ha^{-1}) reduced N_2O emissions by −33% compared to corn stalk alone (12 Mg ha^{-1}) (Hou et al., 2020). Wheat straw (5 Mg ha^{-1}) plus N-fertilizer (225 kg N ha^{-1}) when applied in combination with biochar (8 Mg ha^{-1}) reduced N_2O emissions by −33% compared to only wheat straw plus N-fertilizer (Li et al., 2021). Moreover, application of biochar in combination with compost significantly reduced N_2O emissions by up to −87% (Figure 7c). The reduction in N_2O emissions from biochar-amended compost pile is likely due to increased activity of N_2O-reducing bacteria (Wang et al., 2017) and decreased availability of mineral N to denitrifiers, as biochar can efficiently absorb and retain ammonium (NH_4^+) and nitrate (NO_3^-) ions (Wang et al., 2013). It should be noted that some studies have also reported no effect or increase in N_2O emissions following biochar application (Table 1).

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Such an increase in N_2O emissions was attributed to increased availability of mineral N from the added biochar (Spokas & Reicosky, 2009; Sun et al., 2020) and presence of volatile organic compounds from biochar acting as carbon sources for denitrifiers (Ma, Li Liu, et al., 2019).

The effectiveness of biochar application on soil GHG emissions are affected by various factors such as biochar properties (e.g., feedstock, pyrolysis temperature, C to N ratio, and pH), biochar application rate, soil properties (e.g., soil moisture, texture, and pH), and agronomic activities (e.g., tillage practice, crop rotation, fertilizer, or manure management) (He et al., 2017; Liu, Zhang, et al., 2016; Rittl et al., 2018; Song et al., 2016). The differences in these factors might be responsible for the inconsistent responses of biochar on GHG emissions.

4 | LIMITATIONS AND RESEARCH NEEDS

Despite increased research interest and the large number of publications on the GHG mitigation benefits of biochar, there remain important gaps in our understanding of the extent of these benefits at the field-scale. Addressing these gaps is an urgent research priority to confidently propose the use of biochar application as a land management practice conducive to net C negative agroecosystems. Some of the most important gaps and research priorities identified from this review are summarized in the following.

- Most of the published literatures on biochar and GHG emissions are from laboratory, greenhouse, and mesocosm studies. Due to variable effects of biochar (positive, negative, and no effect) on GHG emissions and the variability in soil, climate, biochar feedstock, pyrolysis temperature and biochar properties, field studies conducted under diverse soil types, climatic conditions and management practices are needed to more fully understand biochar's impact on soil processes and assess its environmental and economic benefits. Additionally, to address the issue of variable impact of biochar, research should be conducted on how biochar manufacturing processes can be tailored for specific conditions (soil, crop, climate, etc.).
- Most of the biochar-related research on GHG emissions has focused on either one or two GHGs (such as CO_2 or N_2O or CH_4) resulting in incomplete assessments of the net climate mitigation benefits of biochar. It is important to evaluate the impact of biochar application on all three major GHGs and estimate GWP for a complete understanding on the contribution of biochar to negative emissions.
- About 89% of the reviewed field studies are short-term (1 to 3 years in duration) and only 11% are medium-term (5 to 7 years). Some studies have reported that the effect of biochar in reducing N_2O emissions can persist for a reasonably long period, as much as 7 years after biochar application. However, this needs to be further explored with long-term field experiments and on-farm assessments integrating both C sequestration and GHG emission reduction in response to biochar applications.
- The lack of long-term studies may be due to the lack of funding, among other factors. Therefore, long-term funding commitment for the establishment and management of these types of studies should be encouraged. Such long-term investigations should provide the information needed to better understand, evaluate, and validate the effect of biochar on soil C residence time, GHG mitigation, and the sustainability of net-zero agricultural systems.
- Emerging research suggests that when (1) manufactured as a biochar-based slow-release fertilizer or (2) combined with manure, compost, crop residue, or N-fertilizer (Figures 6 and 7), biochar is significantly more effective in reducing GHG emissions, mitigating climate change. However, more long-term field research in diverse soil and climate is critical to fully evaluate the suitability of these combinations.
- Biochar application method, including depth (deep vs. shallow vs. surface) of application, is important for farmers managing fields under no-till and minimum tillage. In addition, most of the field studies related to biochar typically involve one-time application. Although one study has indicated a much larger positive effect of split application (compared to one-time application), there is a dearth of information in the literature regarding biochar application frequency. Therefore, more field research (in different soils and climate conditions) is needed to evaluate the environmental and economic benefits of depth and frequency (one-time vs. split) application of biochar.
- Crops have shown diverse responses to different biochar types and application rates. Although few studies have reported 30 Mg ha^{-1} as the optimum rate to significantly reduce GHG emissions while achieving optimum crop yield, that rate seems too high from a cost perspective. Therefore, research should focus on the identification of optimal biochar application rates (maximum yield benefits, GHG reduction, minimum input costs) and time of application in order to improve its adoption level among farmers.
- Despite its numerous benefits, widespread adoption of biochar in farming communities has been hampered by limited product availability, as well as production, application, and transportation costs. Therefore, financial incentives and extension efforts geared toward local production of biochar should help alleviate these constraints. Further, if C abatement and GHG mitigation benefits of biochar application are fully accounted for, the practice should become more

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financially attractive to farmers and that should drive its level of acceptance.
- There is a general lack of awareness, among agricultural producers, of the agronomic and environmental benefits of biochar application to croplands. Therefore, demonstration, educational programs, and national campaign targeting various audiences (farmers, educators, consultants, and planners) are needed to disseminate current knowledge and better manage biochar use in agroecosystems.

5 | CONCLUSIONS

This review primarily focused on the effects of biochar application on GHG emissions, but we also recognize that GHG dynamics in agroecosystems can be complex and impacted by other environmental and land management factors. The information presented in this review has shown that biochar is a technology that can contribute to net-zero emission agricultural systems and thus could help mitigate climate change, one of the most pressing challenges facing humanity in this century. Further, through its additional beneficial impacts on soil health, biochar application may contribute to the sustainability of agricultural production to feed a growing human population without sacrificing critical soil resources. Field research data, collected over the last two decades and reviewed in this paper, have provided ample quantitative evidence of the reduction in GHG emissions that can be achieved through application of biochar to croplands. These findings, along with reports documenting biochar-induced C sequestration improvement, show that agricultural use of biochar may represent an attractive pathway to achieving net C negative agricultural production systems.

AUTHOR CONTRIBUTIONS

Raj K. Shrestha: Conceptualization; Data curation; Formal analysis; Funding acquisition; Methodology; Resources; Visualization; Writing – original draft; Writing – review & editing. **Pierre – Andre Jacinthe**: Conceptualization; Funding acquisition; Supervision; Visualization; Writing – original draft; Writing – review & editing. **Rattan Lal**: Conceptualization; Funding acquisition; Supervision; Visualization; Writing – review & editing. **Klaus Lorenz**: Funding acquisition; Methodology; Resources; Writing – review & editing. **Maninder P. Singh**: Funding acquisition; Validation; Visualization; Writing – review & editing. **Scott M. Demyan**: Funding acquisition; Visualization; Writing – review & editing. **Wei Ren**: Funding acquisition; Writing – original draft; Writing – review & editing. **Laura E. Lindsey**: Conceptualization; Funding acquisition; Methodology; Project administration; Supervision; Visualization; Writing – original draft; Writing – review & editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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REFERENCES

Aamer, M., Hassan, M. U., Shaaban, M., Rasul, F., Haiying, T., Qiaoying, M., Batool, M., Rasheed, A., Chuan, Z., Qitao, S., & Guoqin, H. (2021). Rice straw biochar mitigates N_2O emissions under alternate wetting and drying conditions in paddy soil. *Journal of Saudi Chemical Society*, *25*, 101172. https://doi.org/10.1016/j.jscs.2020.11.005

Abagandura, G. O., Chintala, R., Sandhu, S. S., Kumar, S., & Schumacher, T. E. (2019). Effects of biochar and manure applications on soil carbon dioxide, methane, and nitrous oxide fluxes from two different soils. *Journal of Environmental Quality*, *6*, 1664–1674. https://doi.org/10.2134/jeq2018.10.0374

Agegnehu, G., Bass, A. M., Nelson, P. N., & Bird, M. I. (2016). Benefits of biochar, compost and biochar–compost for soil quality, maize yield and greenhouse gas emissions in a tropical agricultural soil. *Science of the Total Environment*, *543*, 295–306. https://doi.org/10.1016/j.scitotenv.2015.11.054

Agegnehu, G., Srivastava, A. K., & Bird, M. I. (2017). The role of biochar and biochar-compost in improving soil quality and crop performance: A review. *Applied Soil Ecology*, *119*, 156–170. https://doi.org/10.1016/j.apsoil.2017.06.008

Agyarko-Mintah, E., Cowie, A., Singh, B. P., Joseph, S., Van Zwieten, L., Cowie, A., Harden, S., & Smillie, R. (2017). Biochar increases nitrogen retention and lowers greenhouse gas emissions when added to composting poultry litter. *Waste Management*, *61*, 138–149. https://doi.org/10.1016/j.wasman.2016.11.027

Ahmad, M., Rajapaksha, A. U., Lim, J. E., Zhang, M., Bolan, N., Mohan, D., Vithanage, M., Lee, S. S., & Ok, Y. S. (2014). Biochar as a sorbent for contaminant management in soil and water: A review. *Chemosphere*, *99*, 19–33. https://doi.org/10.1016/j.chemosphere.2013.10.071

Aliyu, G., Jiafa, L. U. O., Deyan, L. I. U., Junji, Y. U. A. N., Zengming, C. H. E. N., Tiehu, H. E., & Weixin, D. I. N. G. (2021). Yield-scaled nitrous oxide emissions from nitrogen-fertilized croplands in China: A meta-analysis of contrasting mitigation scenarios. *Pedosphere*, *31*, 231–242. https://doi.org/10.1016/S1002-0160(20)60074-1

Alkharabsheh, H. M., Seleiman, M. F., Battaglia, M. L., Shami, A., Jalal, R. S., Alhammad, B. A., Almutairi, K. F., & Al-Saif, A. M. (2021). Biochar and its broad impacts in soil quality and fertility, nutrient leaching and crop productivity: A review. *Agronomy*, *11*, 993. https://doi.org/10.3390/agronomy11050993

Amoah-Antwi, C., Kwiatkowska-Malina, J., Thornton, S. F., Fenton, O., Malina, G., & Szara, E. (2020). Restoration of soil quality using biochar and brown coal waste: A review. *Science of the Total*

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Environment, *722*, 137852. https://doi.org/10.1016/j.scitotenv.2020.137852

Amonette, J. E., Blanco-Canqui, H., Hassebrook, C., Laird, D. A., Lal, R., Lehmann, J., & Page-Dumroese, D. (2021). Integrated biochar research: A roadmap. *Journal Soil Water Conservation*, *76*, 24A–29A. https://doi.org/10.2489/jswc.2021.1115A

Angst, T. E., Six, J., Reay, D. S., & Sohi, S. P. (2014). Impact of pine chip biochar on trace greenhouse gas emissions and soil nutrient dynamics in an annual ryegrass system in California. *Agriculture, Ecosystems & Environment*, *191*, 17–26. https://doi.org/10.1016/j.agee.2014.03.009

Are, K. S. (2019). Biochar and soil physical health. In V. Abrol, & P. Sharma (Eds.), *Biochar-an imperative amendment for soil and the environment* (pp. 21–33). IntechOpen.

Arif, M., Jan, T., Rasul, F., Riaz, M., Fahad, S., Adnan, M., Amanullah, Ali, K., Mian, I. A., & Khan, B. (2020). Biochar; a remedy for climate change. In S. Fahad, M. Hasanuzzaman, M. Saeed, I. L. Khan, & M. Adnan (Eds.), *Environment, climate, plant and vegetation growth* (pp. 151–171). Springer.

Ashiq, W., & Biswas, A. (2021). Biochar role in mitigation of greenhouse gas emissions from agricultural soils. In A. Rakshit, S. K. Singh, P. C. Abhilash, & A. Biswas (Eds)., *Soil science: Fundamentals to recent advances* (pp. 261–278). Springer.

Ashiq, W., Nadeem, M., Ali, W., Zaeem, M., Wu, J., Galagedara, L., Thomas, R., Kavanagh, V., & Cheema, M. (2020). Biochar amendment mitigates greenhouse gases emission and global warming potential in dairy manure-based silage corn in boreal climate. *Environmental Pollution*, *265*, 114869. https://doi.org/10.1016/j.envpol.2020.114869

Atkinson, C. J., Fitzgerald, J. D., & Hipps, N. A. (2010). Potential mechanisms for achieving agricultural benefits from biochar application to temperate soils: A review. *Plant and Soil*, *337*, 1–18. https://link.springer.com/content/pdf/10.1007/s11104-010-0464-5.pdf

Ayaz, M., Feizienė, D., Tilvikienė, V., Akhtar, K., Stulpinaitė, U., & Iqbal, R. (2021). Biochar role in the sustainability of agriculture and environment. *Sustainability*, *13*, 1330. https://doi.org/10.3390/su13031330

Bach, M., Wilske, B., & Breuer, L. (2016). Current economic obstacles to biochar use in agriculture and climate change mitigation. *Carbon Management*, *7*, 183–190. https://doi.org/10.1080/17583004.2016.1213608

Bai, S. H., Omidvar, N., Gallart, M., Kämper, W., Tahmasbian, I., Farrar, M. B., Singh, K., Zhou, G., Muqadass, B., Xu, C. Y., & Koech, R. (2022). Combined effects of biochar and fertilizer applications on yield: A review and meta-analysis. *Science of the Total Environment*, *808*, 152073. https://doi.org/10.1016/j.scitotenv.2021.152073

Bairwa, R., Mamta, D. L. D., & Bagoria, N. (2020). In-situ trash management induced sustainability of soil health to produce the qualitative products. *Journal Homepage URL*, *5*(2), 249–254.

Barrow, C. J. (2012). Biochar: Potential for countering land degradation and for improving agriculture. *Applied Geography*, *34*, 21–28. https://doi.org/10.1016/j.apgeog.2011.09.008

Basalirwa, D., Sudo, S., Wacal, C., Akae, F., Oo, A. Z., Koyama, S., Sasagawa, D., Yamamoto, S., Masunaga, T., & Nishihara, E. (2020). Assessment of crop residue and palm shell biochar incorporation on greenhouse gas emissions during the fallow and crop growing seasons of broccoli (*Brassica oleracea* var. italica). *Soil and Tillage Research*, *196*, 104435. https://doi.org/10.1016/j.still.2019.104435

Beesley, L., Moreno-Jiménez, E., Gomez-Eyles, J. L., Harris, E., Robinson, B., & Sizmur, T. (2011). A review of biochars' potential role in the remediation, revegetation and restoration of contaminated soils. *Environmental Pollution*, *159*, 3269–3282. https://doi.org/10.1016/j.envpol.2011.07.023

Berihun, T., Tadele, M., & Kebede, F. (2017). The application of biochar on soil acidity and other physico-chemical properties of soils in southern Ethiopia. *Journal of plant nutrition and soil Science*, *180*(3), 381–388. https://doi.org/10.1002/jpln.201600343

Blanco-Canqui, H. (2021). Does biochar improve all soil ecosystem services? *GCB Bioenergy*, *13*, 291–304. https://doi.org/10.1111/gcbb.12783

Borchard, N., Schirrmann, M., Cayuela, M. L., Kammann, C., Wrage-Mönnig, N., Estavillo, J. M., Fuertes-Mendizábal, T., Sigua, G., Spokas, K., Ippolito, J. A., & Novak, J. (2019). Biochar, soil and land-use interactions that reduce nitrate leaching and N_2O emissions: A meta-analysis. *Science of the Total Environment*, *651*, 2354–2364. https://doi.org/10.1016/j.scitotenv.2018.10.060

Cayuela, M. L., Jeffery, S., & van Zwieten, L. (2015). The molar H: Corg ratio of biochar is a key factor in mitigating N_2O emissions from soil. *Agriculture, Ecosystems & Environment*, *202*, 135–138. https://doi.org/10.1016/j.agee.2014.12.015

Cayuela, M. L., Van Zwieten, L., Singh, B. P., Jeffery, S., Roig, A., & Sánchez-Monedero, M. A. (2014). Biochar's role in mitigating soil nitrous oxide emissions: A review and meta-analysis. *Agriculture, Ecosystems & Environment*, *191*, 5–16. https://doi.org/10.1016/j.agee.2013.10.009

Chagas, J. K. M., de Figueiredo, C. C., & Ramos, M. L. G. (2022). Biochar increases soil carbon pools: Evidence from a global meta-analysis. *Journal of Environmental Management*, *305*, 114403. https://doi.org/10.1016/j.jenvman.2021.114403

Chang, J., Clay, D. E., Clay, S. A., Chintala, R., Miller, J. M., & Schumacher, T. (2016). Biochar reduced nitrous oxide and carbon dioxide emissions from soil with different water and temperature cycles. *Agronomy Journal*, *108*, 2214–2221. http://creativecommons.org/licenses/by-nc-nd/4.0/

Charles, A., Rochette, P., Whalen, J. K., Angers, D. A., Chantigny, M. H., & Bertrand, N. (2017). Global nitrous oxide emission factors from agricultural soils after addition of organic amendments: A meta-analysis. *Agriculture, Ecosystems & Environment*, *236*, 88–98. https://doi.org/10.1016/j.agee.2016.11.021

Chen, W., Liao, X., Wu, Y., Liang, J. B., Mi, J., Huang, J., Zhang, H., Wu, Y., Qiao, Z., Li, X., & Wang, Y. (2017). Effects of different types of biochar on methane and ammonia mitigation during layer manure composting. *Waste Management*, *61*, 506–515. https://doi.org/10.1016/j.wasman.2017.01.014

Chun, Y., Lee, S. K., Yoo, H. Y., & Kim, S. W. (2021). Recent advancements in biochar production according to feedstock classification, pyrolysis conditions, and applications: A Review. *BioResources*, *16*(3), 6512–6547. DOI: 10.15376/biores.16.3Chun https://bioresources.cnr.ncsu.edu/wp-content/uploads/2021/07/BioRes_16_3_6512_Chun_LYK_Review_Recent_Biochar_Prodn_Biomass_Classif_Pyrol_18707.pdf

Clough, T. J., Condron, L. M., Kammann, C., & Müller, C. (2013). A review of biochar and soil nitrogen dynamics. *Agronomy*, *3*, 275–293. https://doi.org/10.3390/agronomy3020275

Cong, W., Meng, J., & Ying, S. C. (2018). Impact of soil properties on the soil methane flux response to biochar addition: A meta-analysis.

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Environmental Science: Processes & Impacts, *20*, 1202–1209. https://doi.org/10.1039/C8EM00278A

Dai, Y., Zheng, H., Jiang, Z., & Xing, B. (2020). Combined effects of biochar properties and soil conditions on plant growth: A meta-analysis. *Science of the Total Environment*, *713*, 136635. https://doi.org/10.1016/j.scitotenv.2020.136635

Darmawan, D., Handani, S., & Adli, I. (2021). Improving soil quality by using oil palm solid waste biochar. *AGRITROPICA: Journal of Agricultural Sciences*, *4*(1), 20–29. https://doi.org/10.31186/j.agritropica.4.1.20-29

Das, S. K., Avasthe, R. K., Singh, R., & Babu, S. (2014). Biochar as carbon negative in carbon credit under changing climate. *Current Science*, *107*(7), 1090–1091. https://web.p.ebscohost.com/ehost/pdfviewer/pdfviewer?vid=5&sid=95deb17c-d726-4b36-832a-78da35c9677e%40redis

Dawar, K., Fahad, S., Alam, S. S., Khan, S. A., Dawar, A., Younis, U., Danish, S., Datta, R., & Dick, R. P. (2021). Influence of variable biochar concentration on yield-scaled nitrous oxide emissions, wheat yield and nitrogen use efficiency. *Scientific Reports*, *11*, 1–10. https://doi.org/10.1038/s41598-021-96309-4

Demir, Ö. (2020). Biochar application for greenhouse gases mitigation. In I. Dincer, C. O. Colpan, & M. A. Ezan (Eds.), *Environmentally-benign energy solutions* (pp. 39–68). Springer. https://link.springer.com/content/pdf/10.1007/978-3-030-20637-6.pdf

Deng, B., Fang, H., Jiang, N., Feng, W., Luo, L., Wang, J., Wang, H., Hu, D., Guo, X., & Zhang, L. (2019). Biochar is comparable to dicyandiamide in the mitigation of nitrous oxide emissions from *Camellia oleifera* Abel. Fields. *Forests*, *10*, 1076. https://doi.org/10.3390/f10121076

Dhaliwal, H. S., Brar, Y. S., & Brar, G. S. (2022). Evaluation of pollutants emitted from open field crop residue burning in Punjab, India. In R. Kumar, A. K. Pandey, R. K. Sharma, & G. Norkey (Eds.), *Recent trends in thermal engineering* (pp. 203–211). Springer. https://doi.org/10.1007/978-981-16-3132-0_20

Diatta, A. A., Fike, J. H., Battaglia, M. L., Galbraith, J. M., & Baig, M. B. (2020). Effects of biochar on soil fertility and crop productivity in arid regions: A review. *Arabian Journal of Geosciences*, *13*(14), 1–17. https://doi.org/10.1007/s12517-020-05586-2

Dong, D., Li, J., Ying, S., Wu, J., Han, X., Teng, Y., Zhou, M., Ren, Y., & Jiang, P. (2021). Mitigation of methane emission in a rice paddy field amended with biochar-based slow-release fertilizer. *Science of the Total Environment*, *792*, 148460. https://doi.org/10.1016/j.scitotenv.2021.148460

Egamberdieva, D., Alaylar, B., Kistaubayeva, A., Wirth, S., & Bellingrath-Kimura, S. D. (2022). Biochar for improving soil biological properties and mitigating salt stress in plants on salt-affected soils. *Communications in Soil Science and Plant Analysis*, *53*(2), 140–152. https://doi.org/10.1080/00103624.2021.1993884

Elbasiouny, H., Elbehiry, F., El-Ramady, H., & Hasanuzzaman, M. (2021). Contradictory results of soil greenhouse gas emissions as affected by biochar application: Special focus on alkaline soils. *International Journal of Environmental Research*, *15*(5), 903–920. https://doi.org/10.1007/s41742-021-00358-6

El-Naggar, A., Awad, Y. M., Tang, X. Y., Liu, C., Niazi, N. K., Jien, S. H., Tsang, D. C., Song, H., Ok, Y. S., & Lee, S. S. (2018). Biochar influences soil carbon pools and facilitates interactions with soil: A field investigation. *Land Degradation & Development*, *29*(7), 2162–2171. https://doi.org/10.1002/ldr.2896

El-Naggar, A., Lee, S. S., Rinklebe, J., Farooq, M., Song, H., Sarmah, A. K., Zimmerman, A. R., Ahmad, M., Shaheen, S. M., & Ok, Y. S. (2019). Biochar application to low fertility soils: A review of current status, and future prospects. *Geoderma*, *337*, 536–554. https://doi.org/10.1016/j.geoderma.2018.09.034

Farhangi-Abriz, S., Torabian, S., Qin, R., Noulas, C., Lu, Y., & Gao, S. (2021). Biochar effects on yield of cereal and legume crops using meta-analysis. *Science of the Total Environment*, *775*, 145869. https://doi.org/10.1016/j.scitotenv.2021.145869

Farm Energy. (2019). Biochar: Prospects of *Commercialization*. https://farm-energy.extension.org/biochar-prospects-of-commercialization/ (Accessed 4/12/2022)

Feng, W., Yang, F., Cen, R., Liu, J., Qu, Z., Miao, Q., & Chen, H. (2021). Effects of straw biochar application on soil temperature, available nitrogen and growth of corn. *Journal of Environmental Management*, *277*, 111331. https://doi.org/10.1016/j.jenvman.2020.111331

Feng, Y., Xu, Y., Yu, Y., Xie, Z., & Lin, X. (2012). Mechanisms of biochar decreasing methane emission from Chinese paddy soils. *Soil Biology and Biochemistry*, *46*, 80–88. https://doi.org/10.1016/j.soilbio.2011.11.016

Fidel, R. B., Laird, D. A., & Parkin, T. B. (2019). Effect of biochar on soil greenhouse gas emissions at the laboratory and field scales. *Soil Systems*, *3*(1), 8. https://doi.org/10.3390/soilsystems3010008

Fischer, B. M., Manzoni, S., Morillas, L., Garcia, M., Johnson, M. S., & Lyon, S. W. (2019). Improving agricultural water use efficiency with biochar—A synthesis of biochar effects on water storage and fluxes across scales. *Science of the Total Environment*, *657*, 853–862. https://doi.org/10.1016/j.scitotenv.2018.11.312

Freitas, A. M., Nair, V. D., & Harris, W. G. (2020). Biochar as influenced by feedstock variability: Implications and opportunities for phosphorus management. *Frontiers in Sustainable Food Systems*, *4*, 150. https://doi.org/10.3389/fsufs.2020.510982

Fungo, B., Guerena, D., Thiongo, M., Lehmann, J., Neufeldt, H., & Kalbitz, K. (2014). N_2O and CH_4 emission from soil amended with steam-activated biochar. *Journal of Plant Nutrition and Soil Science*, *177*(1), 34–38. https://doi.org/10.1002/jpln.201300495

Gaunt, J., & Cowie, A. (2012). Biochar, greenhouse gas accounting and emissions trading. In J. Lehmann, & S. Joseph (Eds.), *Biochar for environmental management* (pp. 317–340). Earthscan.

Ghosh, D., & Maiti, S. K. (2021). Biochar-assisted eco-restoration of coal mine degraded land to meet United Nation Sustainable Development Goals. *Land Degradation & Development*, *32*(16), 4494–4508. https://doi.org/10.1002/ldr.4055

Gogoi, N., Sarma, B., Mondal, S. C., Kataki, R., & Garg, A. (2019). Use of biochar in sustainable agriculture. In M. Farooq, & M. Pisante (Eds.), *Innovations in sustainable agriculture* (pp. 501–528). Springer. https://doi.org/10.1007/978-3-030-23169-9

Goswami, M., Pant, G., Mansotra, D. K., Sharma, S., & Joshi, P. C. (2021). Biochar: A carbon negative technology for combating climate change. In D. Pant, A. K. Nadda, K. K. Pant, & A. K. Agarwal (Eds.), *Advances in carbon capture and utilization* (pp. 251–272). Springer. https://doi.org/10.1007/978-981-16-0638-0

Gross, A., Bromm, T., & Glaser, B. (2021). Soil organic carbon sequestration after biochar application: A global meta-analysis. *Agronomy*, *11*(12), 2474. https://doi.org/10.3390/agronomy11122474

Gross, C. D., Bork, E. W., Carlyle, C. N., & Chang, S. X. (2022). Biochar and its manure-based feedstock have divergent effects on soil organic carbon and greenhouse gas emissions in croplands. *Science of the*

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Total Environment, *806*, 151337. https://doi.org/10.1016/j.scitotenv.2021.151337

Guo, M. (2020). The 3R principles for applying biochar to improve soil health. *Soil Systems*, *4*(1), 9. https://doi.org/10.3390/soilsystems4010009

Guo, X. X., Liu, H. T., & Zhang, J. (2020). The role of biochar in organic waste composting and soil improvement: A review. *Waste Management*, *102*, 884–899. https://doi.org/10.1016/j.wasman.2019.12.003

Gupta, D. K., Gupta, C. K., Dubey, R., Fagodiya, R. K., Sharma, G., Keerthika, A., & Shukla, A. K. (2020). Role of biochar in carbon sequestration and greenhouse gas mitigation. In J. S. Singh, & C. Singh (Eds.), *Biochar applications in agriculture and environment management* (pp. 141–165). Springer. https://doi.org/10.1007/978-3-030-40997-5

Hammond, J., Shackley, S., Sohi, S., & Brownsort, P. (2011). Prospective life cycle carbon abatement for pyrolysis biochar systems in the UK. *Energy Policy*, *39*(5), 2646–2655. https://doi.org/10.1016/j.enpol.2011.02.033

Haque, M. M., Rahman, M. M., Morshed, M. M., Islam, M. S., & Afrad, M. S. I. (2019). Biochar on soil fertility and crop productivity. *The Agriculturists*, *17*(1–2), 76–88. https://doi.org/10.3329/agric.v17i1-2.44698

He, C., Manevski, K., Andersen, M. N., Hu, C., Dong, W., & Li, J. (2019). Abiotic mechanisms for biochar effects on soil N_2O emission. *International Agrophysics*, *33*(4), 537–546. doi: 10.31545/intagr/111605

He, T., Yuan, J., Luo, J., Lindsey, S., Xiang, J., Lin, Y., Liu, D., Chen, Z., & Ding, W. (2020). Combined application of biochar with urease and nitrification inhibitors have synergistic effects on mitigating CH_4 emissions in rice field: A three-year study. *Science of the Total Environment*, *743*, 140500. https://doi.org/10.1016/j.scitotenv.2020.140500

He, T., Yun, F., Liu, T., Jin, J., Yang, Y., Fu, Y., & Wang, J. (2021). Differentiated mechanisms of biochar-and straw-induced greenhouse gas emissions in tobacco fields. *Applied Soil Ecology*, *166*, 103996. https://doi.org/10.1016/j.apsoil.2021.103996

He, Y., Zhou, X., Jiang, L., Li, M., Du, Z., Zhou, G., Shao, J., Wang, X., Xu, Z., Hosseini Bai, S., & Xu, C. (2017). Effects of biochar application on soil greenhouse gas fluxes: A meta-analysis. *GCB Bioenergy*, *9*(4), 743–755. https://doi.org/10.1111/gcbb.12376

Hou, R., Li, T., Fu, Q., Liu, D., Li, M., Zhou, Z., Li, Q., Zhao, H., Yu, P., & Yan, J. (2020). Effects of biochar and straw on greenhouse gas emission and its response mechanism in seasonally frozen farmland ecosystems. *Catena*, *194*, 104735. https://doi.org/10.1016/j.catena.2020.104735

Hussain, M., Farooq, M., Nawaz, A., Al-Sadi, A. M., Solaiman, Z. M., Alghamdi, S. S., Ammara, U., Ok, Y. S., & Siddique, K. H. (2017). Biochar for crop production: Potential benefits and risks. *Journal of Soils and Sediments*, *17*(3), 685–716. https://doi.org/10.1007/s11368-016-1360-2

IPCC. (2018). Annex I: Glossary. In V. Masson-Delmotte, P. Zhai, H.-O. Pörtner, D. Roberts, J. Skea, P. R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J. B. R. Matthews, Y. Chen, X. Zhou, M. I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, & T. Waterfield (Eds.), *Global warming of 1.5˚C. An IPCC special report on the impacts of global warming of 1.5˚C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty*. IPCC. https://www.ipcc.ch/site/assets/uploads/sites/2/2019/06/SR15_AnnexI_Glossary.pdf

IPCC. (2019). Summary for policymakers. In P. R. Shukla, J. Skea, E. Calvo Buendia, V. Masson-Delmotte, H.-O. Pörtner, D. C. Roberts, P. Zhai, R. Slade, S. Connors, R. van Diemen, M. Ferrat, E. Haughey, S. Luz, S. Neogi, M. Pathak, J. Petzold, J. Portugal Pereira, P. Vyas, E. Huntley, … K. Kissick (Eds.), *Climate change and land: An IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems*. IPCC. https://www.ipcc.ch/site/assets/uploads/2019/11/SRCCL-Full-Report-Compiled-191128.pdf

Janzen, H. H. (2004). Carbon cycling in earth systems—A soil science perspective. *Agriculture, Ecosystems & Environment*, *104*(3), 399–417. https://doi.org/10.1016/j.agee.2004.01.040

Jeffery, S., Verheijen, F. G., Kammann, C., & Abalos, D. (2016). Biochar effects on methane emissions from soils: A meta-analysis. *Soil Biology and Biochemistry*, *101*, 251–258. https://doi.org/10.1016/j.soilbio.2016.07.021

Jha, P., Biswas, A. K., Lakaria, B. L., & Rao, A. S. (2010). Biochar in agriculture—Prospects and related implications. *Current Science*, *99*(9), 1218–1225. https://www.jstor.org/stable/24068517

Jiang, Z., Lian, F., Wang, Z., & Xing, B. (2020). The role of biochars in sustainable crop production and soil resiliency. *Journal of Experimental Botany*, *71*(2), 520–542. https://doi.org/10.1093/jxb/erz301

Jianwen, Z., Yao, H., Lianggang, Z., Xunhua, Z., & Yuesi, W. (2004). Carbon dioxide, methane, and nitrous oxide emissions from a rice-wheat rotation as affected by crop residue incorporation and temperature. *Advances in Atmospheric Science*, *21*(5), 691–698. https://doi.org/10.1007/BF02916366

Kalu, S., Simojoki, A., Karhu, K., & Tammeorg, P. (2021). Long-term effects of softwood biochar on soil physical properties, greenhouse gas emissions and crop nutrient uptake in two contrasting boreal soils. *Agriculture, Ecosystems & Environment*, *316*, 107454. https://doi.org/10.1016/j.agee.2021.107454

Kammann, C., Ippolito, J., Hagemann, N., Borchard, N., Cayuela, M. L., Estavillo, J. M., Fuertes-Mendizabal, T., Jeffery, S., Kern, J., Novak, J., & Wrage-Mönnig, N. (2017). Biochar as a tool to reduce the agricultural greenhouse-gas burden—Knowns, unknowns and future research needs. *Journal of Environmental Engineering and Landscape Management*, *25*(2), 114–139. https://doi.org/10.3846/16486897.2017.1319375

Kang, S. W., Park, J. W., Seo, D. C., Ok, Y. S., Park, K. D., Choi, I. W., & Cho, J. S. (2016). Effect of biochar application on rice yield and greenhouse gas emission under different nutrient conditions from paddy soil. *Journal of Environmental Engineering*, *142*(10), 04016046. https://doi.org/10.1061/(ASCE)EE.1943-7870.0001083

Kang, S. W., Yun, J. J., Park, J. H., Cheong, Y. H., Park, J. H., Seo, D. C., & Cho, J. S. (2021). Effects of biochar and barley straw application on the rice productivity and greenhouse gas emissions of paddy field. *Applied Biological Chemistry*, *64*(1), 1–10. https://doi.org/10.1186/s13765-021-00666-7

Kapoor, A., Sharma, R., Kumar, A., & Sepehya, S. (2022). Biochar as a means to improve soil fertility and crop productivity: A review. *Journal of Plant Nutrition*, *45*, 2380–2388. https://doi.org/10.1080/01904167.2022.2027980

Karhu, K., Mattila, T., Bergström, I., & Regina, K. (2011). Biochar addition to agricultural soil increased CH4 uptake and water

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

holding capacity—Results from a short-term pilot field study. *Agriculture, Ecosystems & Environment*, *140*(1-2), 309–313. https://doi.org/10.1016/j.agee.2010.12.005

Kavitha, B., Reddy, P. V. L., Kim, B., Lee, S. S., Pandey, S. K., & Kim, K. H. (2018). Benefits and limitations of biochar amendment in agricultural soils: A review. *Journal of Environmental Management*, *227*, 146–154. https://doi.org/10.1016/j.jenvman.2018.08.082

Khan, N., Chowdhary, P., Gnansounou, E., & Chaturvedi, P. (2021). Biochar and environmental sustainability: Emerging trends and techno-economic perspectives. *Bioresource Technology*, *332*, 125102. https://doi.org/10.1016/j.biortech.2021.125102

Kimetu, J. M., & Lehmann, J. (2010). Stability and stabilization of biochar and green manure in soil with different organic carbon contents. *Soil Research*, *48*(7), 577–585. https://doi.org/10.1071/SR10036

Knoblauch, C., Maarifat, A. A., Pfeiffer, E. M., & Haefele, S. M. (2011). Degradability of black carbon and its impact on trace gas fluxes and carbon turnover in paddy soils. *Soil Biology & Biochemistry*, *43*, 1768–1778. https://doi.org/10.1016/j.soilbio.2010.07.012

Kookana, R. S., Sarmah, A. K., Van Zwieten, L., Krull, E., & Singh, B. (2011). Biochar application to soil: Agronomic and environmental benefits and unintended consequences. *Advances in Agronomy*, *112*, 103–143. https://doi.org/10.1016/B978-0-12-385538-1.00003-2

Krause, H. M., Hüppi, R., Leifeld, J., El-Hadidi, M., Harter, J., Kappler, A., Hartmann, M., Behrens, S., Mäder, P., & Gattinger, A. (2018). Biochar affects community composition of nitrous oxide reducers in a field experiment. *Soil Biology and Biochemistry*, *119*, 143–151. https://doi.org/10.1016/j.soilbio.2018.01.018

Kuzyakov, Y., Subbotina, I., Chen, H., Bogomolova, I., & Xu, X. (2009). Black carbon decomposition and incorporation into soil microbial biomass estimated by 14C labeling. *Soil Biology and Biochemistry*, *41*(2), 210–219. https://doi.org/10.1016/j.soilbio.2008.10.016

Laborde, D., Mamun, A., Martin, W., Piñeiro, V., & Vos, R. (2021). Agricultural subsidies and global greenhouse gas emissions. *Nature Communications*, *12*(1), 1–9. https://doi.org/10.1038/s41467-021-22703-1

Lal, R. (2021). Negative emission farming. *Journal of Soil and Water Conservation*, *76*(3), 61A–64A. https://doi.org/10.2489/jswc.2021.0419A

Latawiec, A. E., Królczyk, J. B., Kuboń, M., Szwedziak, K., Drosik, A., Polańczyk, E., Grotkiewicz, K., & Strassburg, B. B. (2017). Willingness to adopt biochar in agriculture: The producer's perspective. *Sustainability*, *9*(4), 655. https://doi.org/10.3390/su9040655

Lee, J. W., Hawkins, B., Day, D. M., & Reicosky, D. C. (2010). Sustainability: The capacity of smokeless biomass pyrolysis for energy production, global carbon capture and sequestration. *Energy & Environmental Science*, *3*(11), 1695–1705. https://doi.org/10.1039/C004561F

Lehmann, J., Cowie, A., Masiello, C. A., Kammann, C., Woolf, D., Amonette, J. E., Cayuela, M. L., Camps-Arbestain, M., & Whitman, T. (2021). Biochar in climate change mitigation. *Nature Geoscience*, *14*(12), 883–892. https://doi.org/10.1038/s41561-021-00852-8

Lehmann, J., Gaunt, J., & Rondon, M. (2006). Bio-char sequestration in terrestrial ecosystems—A review. *Mitigation and Adaptation Strategies for Global Change*, *11*(2), 403–427. https://doi.org/10.1007/s11027-005-9006-5

Lehmann, J., & Joseph, S. (2015). Biochar for environmental management: An introduction. In J. Lehmann, & S. Joseph (Eds.), *Biochar for environmental management* (pp. 33–46). Routledge. https://doi.org/10.4324/9780203762264

Lehmann, J., Rillig, M. C., Thies, J., Masiello, C. A., Hockaday, W. C., & Crowley, D. (2011). Biochar effects on soil biota—A review. *Soil Biology and Biochemistry*, *43*(9), 1812–1836. https://doi.org/10.1016/j.soilbio.2011.04.022

Leng, L., Xiong, Q., Yang, L., Li, H., Zhou, Y., Zhang, W., Jiang, S., Li, H., & Huang, H. (2021). An overview on engineering the surface area and porosity of biochar. *Science of the total Environment*, *763*, 144204. https://doi.org/10.1016/j.scitotenv.2020.144204

Lentz, R. D., Ippolito, J. A., & Spokas, K. A. (2014). Biochar and manure effects on net nitrogen mineralization and greenhouse gas emissions from calcareous soil under corn. *Soil Science Society of America Journal*, *78*, 1641–1655. https://doi.org/10.2136/sssaj2014.05.0198

Li, B., Fan, C. H., Zhang, H., Chen, Z. Z., Sun, L. Y., & Xiong, Z. Q. (2015). Combined effects of nitrogen fertilization and biochar on the net global warming potential, greenhouse gas intensity and net ecosystem economic budget in intensive vegetable agriculture in southeastern China. *Atmospheric Environment*, *100*, 10–19. https://doi.org/10.1016/j.atmosenv.2014.10.034

Li, F., Duan, F., Ji, W., & Gui, X. (2020). Biochar-activated persulfate for organic contaminants removal: Efficiency, mechanisms and influencing factors. *Ecotoxicology and Environmental Safety*, *198*, 110653. https://doi.org/10.1016/j.ecoenv.2020.110653

Li, N., Wen, S., Wei, S., Li, H., Feng, Y., Ren, G., Yang, G., Han, X., Wang, X., & Ren, C. (2021). Straw incorporation plus biochar addition improved the soil quality index focused on enhancing crop yield and alleviating global warming potential. *Environmental Technology & Innovation*, *21*, 101316. https://doi.org/10.1016/j.eti.2020.101316

Liang, C., Amelung, W., Lehmann, J., & Kästner, M. (2019). Quantitative assessment of microbial necromass contribution to soil organic matter. *Global Change Biology*, *25*(11), 3578–3590. https://doi.org/10.1111/gcb.14781

Liu, L., & Greaver, T. L. (2009). A review of nitrogen enrichment effects on three biogenic GHGs: The CO2 sink may be largely offset by stimulated N_2O and CH_4 emission. *Ecology Letters*, *12*(10), 1103–1117. https://doi.org/10.1111/j.1461-0248.2009.01351.x

Liu, L., Shen, G., Sun, M., Cao, X., Shang, G., & Chen, P. (2014). Effect of biochar on nitrous oxide emission and its potential mechanisms. *Journal of the Air & Waste Management Association*, *64*(8), 894–902. https://doi.org/10.1080/10962247.2014.899937

Liu, M., Almatrafi, E., Zhang, Y., Xu, P., Song, B., Zhou, C., Zeng, G., & Zhu, Y. (2022). A critical review of biochar-based materials for the remediation of heavy metal contaminated environment: Applications and practical evaluations. *Science of the Total Environment*, *806*, 150531. https://doi.org/10.1016/j.scitotenv.2021.150531

Liu, Q., Liu, B., Ambus, P., Zhang, Y., Hansen, V., Lin, Z., Shen, D., Liu, G., Bei, Q., Zhu, J., & Wang, X. (2016). Carbon footprint of rice production under biochar amendment—A case study in a Chinese rice cropping system. *GCB Bioenergy*, *8*(1), 148–159. https://doi.org/10.1111/gcbb.12248

Liu, S., Zhang, Y., Zong, Y., Hu, Z., Wu, S., Zhou, J., Jin, Y., & Zou, J. (2016). Response of soil carbon dioxide fluxes, soil organic carbon and microbial biomass carbon to biochar amendment: A meta-analysis. *GCB Bioenergy*, *8*(2), 392–406. https://doi.org/10.1111/gcbb.12265

Liu, X., Mao, P., Li, L., & Ma, J. (2019). Impact of biochar application on yield-scaled greenhouse gas intensity: A meta-analysis. *Science of the*

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Total Environment, *656*, 969–976. https://doi.org/10.1016/j.scitotenv.2018.11.396

Liu, X., Zhou, J., Chi, Z., Zheng, J., Li, L., Zhang, X., Zheng, J., Cheng, K., Bian, R., & Pan, G. (2019). Biochar provided limited benefits for rice yield and greenhouse gas mitigation six years following an amendment in a fertile rice paddy. *Catena*, *179*, 20–28. https://doi.org/10.1016/j.catena.2019.03.033

Liu, Y., Bi, Y., Xie, Y., Zhao, X., He, D., Wang, S., Wang, C., Guo, T., & Xing, G. (2020). Successive straw biochar amendments reduce nitrous oxide emissions but do not improve the net ecosystem economic benefit in an alkaline sandy loam under a wheat–maize cropping system. *Land Degradation & Development*, *31*(7), 868–883. https://doi.org/10.1002/ldr.3495

Lu, X., Li, Y., Wang, H., Singh, B. P., Hu, S., Luo, Y., Li, J., Xiao, Y., Cai, X., & Li, Y. (2019). Responses of soil greenhouse gas emissions to different application rates of biochar in a subtropical Chinese chestnut plantation. *Agricultural and Forest Meteorology*, *271*, 168–179. https://doi.org/10.1016/j.agrformet.2019.03.001

Lu, Y., Silveira, M. L., Cavigelli, M., O'Connor, G. A., Vendramini, J. M., Erickson, J. E., & Li, Y. C. (2020). Biochar impacts on nutrient dynamics in a subtropical grassland soil: 2. Greenhouse gas emissions. *Journal of Environmental Quality*, *49*, 1421–1434. https://doi.org/10.1002/jeq2.20141

Lynch, J., Cain, M., Frame, D., & Pierrehumbert, R. (2021). Agriculture's contribution to climate change and role in mitigation is distinct from predominantly fossil CO_2-emitting sectors. *Frontiers in Sustainable Food Systems*, *4*, 518039. https://doi.org/10.3389/fsufs.2020.518039

Ma, H., Egamberdieva, D., Wirth, S., & Bellingrath-Kimura, S. D. (2019). Effect of biochar and irrigation on soybean-rhizobium symbiotic performance and soil enzymatic activity in field rhizosphere. *Agronomy*, *9*(10), 626. https://doi.org/10.3390/agronomy9100626

Ma, L., Kong, F., Lv, X., Wang, Z., Zhou, Z., & Meng, Y. (2021). Responses of greenhouse gas emissions to different straw management methods with the same amount of carbon input in cotton field. *Soil and Tillage Research*, *213*, 105126. https://doi.org/10.1016/j.still.2021.105126

Ma, Y., Li Liu, D., Schwenke, G., & Yang, B. (2019). The global warming potential of straw-return can be reduced by application of straw-decomposing microbial inoculants and biochar in rice-wheat production systems. *Environmental Pollution*, *252*, 835–845. https://doi.org/10.1016/j.envpol.2019.06.006

Major, J., Lehmann, J., Rondon, M., & Goodale, C. (2010). Fate of soil-applied black carbon: Downward migration, leaching and soil respiration. *Global Change Biology*, *16*(4), 1366–1379. https://doi.org/10.1111/j.1365-2486.2009.02044.x

Major, J., Steiner, C., Downie, A., & Lehmann, J. (2009). Biochar effects on nutrient leaching. In J. Lehmann, & S. Joseph (Eds.), *Biochar for environmental management: Science and technology* (pp. 271–288). Earthscan.

Majumder, S., Neogi, S., Dutta, T., Powel, M. A., & Banik, P. (2019). The impact of biochar on soil carbon sequestration: Meta-analytical approach to evaluating environmental and economic advantages. *Journal of Environmental Management*, *250*, 109466. https://doi.org/10.1016/j.jenvman.2019.109466

Man, Y., Wang, B., Wang, J., Slaný, M., Yan, H., Li, P., El-Naggar, A., Shaheen, S. M., Rinklebe, J., & Feng, X. (2021). Use of biochar to reduce mercury accumulation in *Oryza sativa* L.: A trial for sustainable management of historically polluted farmlands. *Environment International*, *153*, 106527. https://doi.org/10.1016/j.envint.2021.106527

Matuštík, J., Hnátková, T., & Kočí, V. (2020). Life cycle assessment of biochar-to-soil systems: A review. *Journal of Cleaner Production*, *259*, 120998. https://doi.org/10.1016/j.jclepro.2020.120998

Mer, J. L., & Roger, P. (2001). Production, oxidation, emission and consumption of methane by soils: A review. *European Journal of Soil Biology*, *37*(1), 25–50. https://doi.org/10.1016/S1164-5563(01)01067-6

Mishra, B., Aggarwal, M., & Remya, N. (2022). Application of biochar for removal of emerging contaminants. In S. P. Singh, A. K. Agarwal, T. Gupta, & S. M. Maliyekkal (Eds.), *New trends in emerging environmental contaminants* (pp. 211–224). Springer. https://doi.org/10.1007/978-981-16-8367-1_10

Mohan, D., Sarswat, A., Ok, Y. S., & Pittman, C. U. Jr. (2014). Organic and inorganic contaminants removal from water with biochar, a renewable, low cost and sustainable adsorbent—A critical review. *Bioresource Technology*, *160*, 191–202. https://doi.org/10.1016/j.biortech.2014.01.120

Mohawesh, O., Albalasmeh, A., Gharaibeh, M., Deb, S., Simpson, C., Singh, S., Al-Soub, B., & Hanandeh, A. E. (2021). Potential use of biochar as an amendment to improve soil fertility and tomato and bell pepper growth performance under arid conditions. *Journal of Soil Science and Plant Nutrition*, *21*(4), 2946–2956. https://doi.org/10.1007/s42729-021-00580-3

Mosharrof, M., Uddin, M., Jusop, S., Sulaiman, M. F., Shamsuzzaman, S. M., & Haque, A. N. (2021). Integrated use of biochar and lime as a tool to improve maize yield and mitigate CO_2 emission: A review. *Chilean Journal of Agricultural Research*, *81*(1), 109–118. https://doi.org/10.4067/S0718-58392021000100109

Mosier, A. R., Duxbury, J. M., Freney, J. R., Heinemeyer, O., Minami, K., & Johnson, D. E. (1998). Mitigating agricultural emissions of methane. *Climatic Change*, *40*(1), 39–80. https://doi.org/10.1023/A:1005338731269

Mukherjee, A., & Lal, R. (2013). Biochar impacts on soil physical properties and greenhouse gas emissions. *Agronomy*, *3*(2), 313–339. https://doi.org/10.3390/agronomy3020313

Nan, Q., Wang, C., Wang, H., Yi, Q., & Wu, W. (2020). Mitigating methane emission via annual biochar amendment pyrolyzed with rice straw from the same paddy field. *Science of the Total Environment*, *746*, 141351. https://doi.org/10.1016/j.scitotenv.2020.141351

Nan, Q., Xin, L., Qin, Y., Waqas, M., & Wu, W. (2021). Exploring long-term effects of biochar on mitigating methane emissions from paddy soil: A review. *Biochar*, *3*(2), 125–134. https://doi.org/10.1007/s42773-021-00096-0

NOAA. (2022). *Trends in atmospheric methane. Global monitoring laboratory*. National Oceanic and Atmospheric Administration. https://gml.noaa.gov/ccgg/ (Accessed 2/4/2022)

O'Connor, D., Peng, T., Zhang, J., Tsang, D. C., Alessi, D. S., Shen, Z., Bolan, N. S., & Hou, D. (2018). Biochar application for the remediation of heavy metal polluted land: A review of in situ field trials. *Science of the Total Environment*, *619*, 815–826. https://doi.org/10.1016/j.scitotenv.2017.11.132

O'Connor, J. J., Fest, B. J., Sievers, M., & Swearer, S. E. (2020). Impacts of land management practices on blue carbon stocks and greenhouse gas fluxes in coastal ecosystems—A meta-analysis. *Global Change Biology*, *26*(3), 1354–1366. https://doi.org/10.1111/gcb.14946

Oladele, S. O. (2019). Changes in physicochemical properties and quality index of an Alfisol after three years of rice husk biochar amendment

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in rainfed rice–maize cropping sequence. *Geoderma*, *353*, 359–371. https://doi.org/10.1016/j.geoderma.2019.06.038

Oladele, S. O., & Adetunji, A. T. (2021). Agro-residue biochar and N fertilizer addition mitigates CO_2-C emission and stabilized soil organic carbon pools in a rain-fed agricultural cropland. *International Soil and Water Conservation Research*, *9*(1), 76–86. https://doi.org/10.1016/j.iswcr.2020.09.002

Olszyk, D., Shiroyama, T., Novak, J., Cantrell, K., Sigua, G., Watts, D., & Johnson, M. G. (2020). Biochar affects growth and shoot nitrogen in four crops for two soils. *Agrosystems, Geosciences & Environment*, *3*(1), e20067. https://doi.org/10.1002/agg2.20067

Ong, S. H., Tan, R. R., & Andiappan, V. (2021). Optimisation of biochar-based supply chains for negative emissions and resource savings in carbon management networks. *Clean Technologies and Environmental Policy*, *23*(2), 621–638. https://doi.org/10.1007/s10098-020-01990-0

Oni, B. A., Oziegbe, O., & Olawole, O. O. (2019). Significance of biochar application to the environment and economy. *Annals of Agricultural Sciences*, *64*(2), 222–236. https://doi.org/10.1016/j.aoas.2019.12.006

Oregon Biochar Solutions. (2022). Rogue *B*iochar *P*ricing. https://www.chardirect.com/rogue-biochar-pricing (Accessed 4/12/2022)

Palai, J. B. (2020). Role of biochar in sustainable agriculture. *Agro Economist*, *7*, 73–75.

Panchasara, H., Samrat, N. H., & Islam, N. (2021). Greenhouse gas emissions trends and mitigation measures in Australian agriculture sector—A review. *Agriculture*, *11*(2), 85. https://doi.org/10.3390/agriculture11020085

Platzer, M. F., & Sarigul-Klijn, N. (2021). Climate tipping points and climate irreversibility. In M. F. Platzer, & N. Sarigul-Klijn (Eds.), *The green energy ship concept* (pp. 17–20). Springer.

Purakayastha, T. J., Bera, T., Bhaduri, D., Sarkar, B., Mandal, S., Wade, P., Kumari, S., Biswas, S., Menon, M., Pathak, H., & Tsang, D. C. (2019). A review on biochar modulated soil condition improvements and nutrient dynamics concerning crop yields: Pathways to climate change mitigation and global food security. *Chemosphere*, *227*, 345–365. https://doi.org/10.1016/j.chemosphere.2019.03.170

Purakayastha, T. J., Bhaduri, D., & Singh, P. (2021). Role of biochar on greenhouse gas emissions and carbon sequestration in soil: Opportunities for mitigating climate change. In A. Rakshit, S. K. Singh, P. C. Abhilash, & A. Biswas (Eds.), *Soil science: Fundamentals to recent advances* (pp. 237–260). Springer.

Qi, L., Pokharel, P., Chang, S. X., Zhou, P., Niu, H., He, X., Wang, Z., & Gao, M. (2020). Biochar application increased methane emission, soil carbon storage and net ecosystem carbon budget in a 2-year vegetable–rice rotation. *Agriculture, Ecosystems & Environment*, *292*, 106831. https://doi.org/10.1016/j.agee.2020.106831

Ravindra, K., Singh, T., & Mor, S. (2019). Emissions of air pollutants from primary crop residue burning in India and their mitigation strategies for cleaner emissions. *Journal of Cleaner Production*, *208*, 261–273. https://doi.org/10.1016/j.jclepro.2018.10.031

Razzaghi, F., Obour, P. B., & Arthur, E. (2020). Does biochar improve soil water retention? A systematic review and meta-analysis. *Geoderma*, *361*, 114055. https://doi.org/10.1016/j.geoderma.2019.114055

Risk, N., Snider, D., & Wagner-Riddle, C. (2013). Mechanisms leading to enhanced soil nitrous oxide fluxes induced by freeze–thaw cycles. *Canadian Journal of Soil Science*, *93*(4), 401–414. https://doi.org/10.4141/cjss2012-071

Ritchie, H., & Roser, M. (2022). CO2 and GHG Emissions by Sector. Our World in Data. https://ourworldindata.org/emissions-by-sector (Accessed 2/7/2022)

Rittl, T. F., Butterbach-Bahl, K., Basile, C. M., Pereira, L. A., Alms, V., Dannenmann, M., Couto, E. G., & Cerri, C. E. (2018). Greenhouse gas emissions from soil amended with agricultural residue biochars: Effects of feedstock type, production temperature and soil moisture. *Biomass and Bioenergy*, *117*, 1–9. https://doi.org/10.1016/j.biombioe.2018.07.004

Rittl, T. F., Oliveira, D., Canisares, L. P., Sagrilo, E., Butterbach-Bahl, K., Dannenmann, M., & Cerri, C. E. P. (2021). High application rates of biochar to mitigate N_2O emissions from a N-fertilized tropical soil under warming conditions. *Frontiers in Environmental Science*, *8*, 1–5. https://doi.org/10.3389/fenvs.2020.611873

Roberts, K. G., Gloy, B. A., Joseph, S., Scott, N. R., & Lehmann, J. (2010). Life cycle assessment of biochar systems: Estimating the energetic, economic, and climate change potential. *Environmental Science & Technology*, *44*(2), 827–833. https://doi.org/10.1021/es902266r

Rodriguez-Franco, C., & Page-Dumroese, D. S. (2021). Woody biochar potential for abandoned mine land restoration in the US: A review. *Biochar*, *3*(1), 7–22. https://doi.org/10.1007/s42773-020-00074-y

Safaei, K. M., Zhang, G., Akram, F., Rudolf, K., Adeel, M., Sasan, J., Pauzi, Z. M., & Li, G. (2020). Effect of walnut shell biochars on soil quality, crop yields, and weed dynamics in a 4-year field experiment. *Environmental Science and Pollution Research International*, *27*(15), 18510–18520. https://doi.org/10.1007/s11356-020-08335-w

Schmidt, H. P., Kammann, C., Hagemann, N., Leifeld, J., Bucheli, T. D., Sánchez Monedero, M. A., & Cayuela, M. L. (2021). Biochar in agriculture—A systematic review of 26 global meta-analyses. *GCB Bioenergy*, *13*(11), 1708–1730. https://doi.org/10.1111/gcbb.12889

Seki, M., Sugihara, S., Miyazaki, H., Jegadeesan, M., Kannan, P., Bertrand, I., & Tanaka, H. (2022). Impact of biochar and manure application on in-situ carbon dioxide flux, microbial activity, and carbon budget in degraded cropland soil of southern India. *Land Degradation & Development*, *33*(10), 1626–1636. https://doi.org/10.1002/ldr.4234

Semida, W. M., Beheiry, H. R., Sétamou, M., Simpson, C. R., Abd El-Mageed, T. A., Rady, M. M., & Nelson, S. D. (2019). Biochar implications for sustainable agriculture and environment: A review. *South African Journal of Botany*, *127*, 333–347. https://doi.org/10.1016/j.sajb.2019.11.015

Shakoor, A., Arif, M. S., Shahzad, S. M., Farooq, T. H., Ashraf, F., Altaf, M. M., Ahmed, W., Tufail, M. A., & Ashraf, M. (2021). Does biochar accelerate the mitigation of greenhouse gaseous emissions from agricultural soil? – A global meta-analysis. *Environmental Research*, *202*, 111789. https://doi.org/10.1016/j.envres.2021.111789

Shakoor, A., Shahzad, S. M., Chatterjee, N., Arif, M. S., Farooq, T. H., Altaf, M. M., Tufail, M. A., Dar, A. A., & Mehmood, T. (2021). Nitrous oxide emission from agricultural soils: Application of animal manure or biochar? A global meta-analysis. *Journal of Environmental Management*, *285*, 112170. https://doi.org/10.1016/j.jenvman.2021.112170

Shen, J., Tang, H., Liu, J., Wang, C., Li, Y., Ge, T., Jones, D. L., & Wu, J. (2014). Contrasting effects of straw and straw-derived biochar amendments on greenhouse gas emissions within double rice cropping systems. *Agriculture, Ecosystems & Environment*, *188*, 264–274. https://doi.org/10.1016/j.agee.2014.03.002

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Shen, Y., Zhu, L., Cheng, H., Yue, S., & Li, S. (2017). Effects of biochar application on CO_2 emissions from a cultivated soil under semiarid climate conditions in northwest China. *Sustainability*, *9*(8), 1482. https://doi.org/10.3390/su9081482

Singh, B. P., Hatton, B. J., Singh, B., Cowie, A. L., & Kathuria, A. (2010). Influence of biochars on nitrous oxide emission and nitrogen leaching from two contrasting soils. *Journal of Environmental Quality*, *39*(4), 1224–1235. https://doi.org/10.2134/jeq2009.0138

Sistani, K. R., Simmons, J. R., Jn-Baptiste, M., & Novak, J. M. (2019). Poultry litter, biochar, and fertilizer effect on corn yield, nutrient uptake, N_2O and CO_2 emissions. *Environments*, *6*(5), 55. https://doi.org/10.3390/environments6050055

Smith, J. L., Collins, H. P., & Bailey, V. L. (2010). The effect of young biochar on soil respiration. *Soil Biology and Biochemistry*, *42*(12), 2345–2347. https://doi.org/10.1016/j.soilbio.2010.09.013

Smith, P. (2016). Soil carbon sequestration and biochar as negative emission technologies. *Global Change Biology*, *22*(3), 1315–1324. https://doi.org/10.1111/gcb.13178

Smith, P., Bustamante, M., Ahammad, H., Clark, H., Dong, H., Elsiddig, E. A., Haberl, H., Harper, R., House, J., Jafari, M., Masera, O., Mbow, C., Ravindranath, N. H., Rice, C. W., Robledo Abad, C., Romanovskaya, A., Sperling, F., & Tubiello, F. (2014). Agriculture, forestry and other land sse (AFOLU). In O. Edenhofer, R. Pichs-Madruga, Y. Sokona, E. Farahani, S. Kadner, P. Seyboth, K. Adler, A. Baum, I. Brunner, S. Eickemeier, B. Kriemann, J. Savolainen, S. Schlömer, C. Stechow, T. von, Zwickel, & J. C. Minx (Eds.), *Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Climate change 2014: Mitigation of climate change* (pp. 811–922). Cambridge University Press. https://www.ipcc.ch/site/assets/uploads/2018/02/ipcc_wg3_ar5_full.pdf

Sohi, S. P., Krull, E., Lopez-Capel, E., & Bol, R. (2010). A review of biochar and its use and function in soil. *Advances in Agronomy*, *105*, 47–82. https://doi.org/10.1016/S0065-2113(10)05002-9

Song, K., Zhang, G., Ma, J., Peng, S., Lv, S., & Xu, H. (2022). Greenhouse gas emissions from ratoon rice fields among different varieties. *Field Crops Research*, *277*, 108423. https://doi.org/10.1016/j.fcr.2021.108423

Song, X., Pan, G., Zhang, C., Zhang, L., & Wang, H. (2016). Effects of biochar application on fluxes of three biogenic greenhouse gases: A meta-analysis. *Ecosystem Health and Sustainability*, *2*(2), e01202. https://doi.org/10.1002/ehs2.1202

Sonoki, T., Furukawa, T., Jindo, K., Suto, K., Aoyama, M., & Sánchez-Monedero, M. Á. (2013). Influence of biochar addition on methane metabolism during thermophilic phase of composting. *Journal of Basic Microbiology*, *53*(7), 617–621. https://doi.org/10.1002/jobm.201200096

Spokas, K. A., Baker, J. M., & Reicosky, D. C. (2010). Ethylene: Potential key for biochar amendment impacts. *Plant and Soil*, *333*(1), 443–452. https://doi.org/10.1007/s11104-010-0359-5

Spokas, K. A., Cantrell, K. B., Novak, J. M., Archer, D. W., Ippolito, J. A., Collins, H. P., Boateng, A. A., Lima, I. M., Lamb, M. C., McAloon, A. J., & Nichols, K. A. (2012). Biochar: A synthesis of its agronomic impact beyond carbon sequestration. *Journal of Environmental Quality*, *41*(4), 973–989. https://doi.org/10.2134/jeq2011.0069

Spokas, K. A., & Reicosky, D. C. (2009). Impacts of sixteen different biochars on soil greenhouse gas production. *Annals of Environmental Science*, *3*, 179–193. https://openjournals.neu.edu/aes/journal/article/view/v3art4/v3p179-193

Sriphirom, P., Chidthaisong, A., Yagi, K., Tripetchkul, S., Boonapatcharoen, N., & Towprayoon, S. (2021). Effects of biochar on methane emission, grain yield, and soil in rice cultivation in Thailand. *Carbon Management*, *12*(2), 109–121. https://doi.org/10.1080/17583004.2021.1885257

Sriphirom, P., Chidthaisong, A., Yagi, K., Tripetchkul, S., & Towprayoon, S. (2020). Evaluation of biochar applications combined with alternate wetting and drying (AWD) water management in rice field as a methane mitigation option for farmers' adoption. *Soil Science and Plant Nutrition*, *66*(1), 235–246. https://doi.org/10.1080/00380768.2019.1706431

Sri Shalini, S., Palanivelu, K., Ramachandran, A., & Raghavan, V. (2021). Biochar from biomass waste as a renewable carbon material for climate change mitigation in reducing greenhouse gas emissions—A review. *Biomass Conversion and Biorefinery*, *11*(5), 2247–2267. https://doi.org/10.1007/s13399-020-00604-5

Stavi, I. (2012). The potential use of biochar in reclaiming degraded rangelands. *Journal of Environmental Planning and Management*, *55*(5), 657–665. https://doi.org/10.1080/09640568.2011.620333

Sui, Y., Gao, J., Liu, C., Zhang, W., Lan, Y., Li, S., Meng, J., Xu, Z., & Tang, L. (2016). Interactive effects of straw-derived biochar and N fertilization on soil C storage and rice productivity in rice paddies of northeast China. *Science of the Total Environment*, *544*, 203–210. https://doi.org/10.1016/j.scitotenv.2015.11.079

Sun, L., Deng, J., Fan, C., Li, J., & Liu, Y. (2020). Combined effects of nitrogen fertilizer and biochar on greenhouse gas emissions and net ecosystem economic budget from a coastal saline rice field in southeastern China. *Environmental Science and Pollution Research*, *27*(14), 17013–17022. https://doi.org/10.1007/s11356-020-08204-6

Sun, Y., Xiong, X., He, M., Xu, Z., Hou, D., Zhang, W., Ok, Y. S., Rinklebe, J., Wang, L., & Tsang, D. C. (2021). Roles of biochar-derived dissolved organic matter in soil amendment and environmental remediation: A critical review. *Chemical Engineering Journal*, *424*, 130387. https://doi.org/10.1016/j.cej.2021.130387

Tan, G., Wang, H., Xu, N., Junaid, M., Liu, H., & Zhai, L. (2021). Effects of biochar application with fertilizer on soil microbial biomass and greenhouse gas emissions in a peanut cropping system. *Environmental Technology*, *42*, 9–19. https://doi.org/10.1080/09593330.2019.1620344

Tang, Y., Gao, W., Cai, K., Chen, Y., Li, C., Lee, X., Cheng, H., Zhang, Q., & Cheng, J. (2021). Effects of biochar amendment on soil carbon dioxide emission and carbon budget in the karst region of southwest China. *Geoderma*, *385*, 114895. https://doi.org/10.1016/j.geoderma.2020.114895

Terra Global Capital. (2010). Evaluation of the Opportunities for Generating Carbon Offsets from Soil Sequestration of Biochar: An Issues Paper Commissioned by the Climate Action Reserve. https://www.climateactionreserve.org/wp-content/uploads/2009/03/Soil_Sequestration_Biochar_Issue_Paper1.pdf

The Climate Trust. (2010). Carbon Markets Investment Criteria for Biochar Project. The Carbon Trust, September 2010. Public Interest Energy Research (PIER) Program. Final Project Report. CEC-500-02-004. https://climatetrust.org/wp-content/uploads/2014/07/WESTCARB-Report-Carbon-Market-Investment-Criteria-for-Biochar-Projects.pdf

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Thers, H., Abalos, D., Dörsch, P., & Elsgaard, L. (2020). Nitrous oxide emissions from oilseed rape cultivation were unaffected by flash pyrolysis biochar of different type, rate and field ageing. *Science of the Total Environment*, *724*, 138140. https://doi.org/10.1016/j.scitotenv.2020.138140

Tisserant, A., & Cherubini, F. (2019). Potentials, limitations, co-benefits, and trade-offs of biochar applications to soils for climate change mitigation. *Land*, *8*(12), 179. https://doi.org/10.3390/land8120179

Troy, S. M., Lawlor, P. G., Flynn, C. J. O., & Healy, M. G. (2013). Impact of biochar addition to soil on greenhouse gas emissions following pig manure application. *Soil Biology and Biochemistry*, *60*, 173–181.

UNFCCC. (2015). *Paris Agreement. United Nations Framework Convention on Climate Change*. https://unfccc.int/sites/default/files/english_paris_agreement.pdf

Van Zwieten, L., Kimber, S., Morris, S., Downie, A., Berger, E., Rust, J., & Scheer, C. (2010). Influence of biochars on flux of N_2O and CO_2 from Ferrosol. *Soil Research*, *48*(7), 555–568. https://doi.org/10.1071/SR10004

Van Zwieten, L., Rose, T., Herridge, D., Kimber, S., Rust, J., Cowie, A., & Morris, S. (2015). Enhanced biological N_2 fixation and yield of faba bean (*Vicia faba* L.) in an acid soil following biochar addition: Dissection of causal mechanisms. *Plant and Soil*, *395*(1), 7–20. https://doi.org/10.1007/s11104-015-2427-3

Van Zwieten, L., Singh, B., Joseph, S., Kimber, S., Cowie, A., & Chan, K. Y. (2009). Biochar and emissions of non-CO_2 greenhouse gases from soil. In J. Lehmann, & S. Joseph (Eds.), *Biochar for environmental management – Science and technology* (pp. 227–249). Earthscan.

Van Zwieten, L., Singh, B. P., Kimber, S. W. L., Murphy, D. V., Macdonald, L. M., Rust, J., & Morris, S. (2014). An incubation study investigating the mechanisms that impact N_2O flux from soil following biochar application. *Agriculture, Ecosystems & Environment*, *191*, 53–62. https://doi.org/10.1016/j.agee.2014.02.030

Van Zwieten, L., Singh, B. P., Tavakkoli, E., Joseph, S., Macdonald, L. M., Rose, T. J., Rose, M. T., Kimber, S. W., Morris, S., Cozzolino, D., & Araujo, J. R. (2017). Biochar built soil carbon over a decade by stabilizing rhizodeposits. *Nature Climate Change*, *7*(5), 371–376. https://doi.org/10.1038/nclimate3276

Verhoeven, E., Pereira, E., Decock, C., Suddick, E., Angst, T., & Six, J. (2017). Toward a better assessment of biochar–nitrous oxide mitigation potential at the field scale. *Journal of Environmental Quality*, *46*(2), 237–246. https://doi.org/10.2134/jeq2016.10.0396

Wakefield Biochar. (2021). Wakefield Biochar. https://www.wakefieldbiochar.com/

Wang, C., Lu, H., Dong, D., Deng, H., Strong, P. J., Wang, H., & Wu, W. (2013). Insight into the effects of biochar on manure composting: Evidence supporting the relationship between N_2O emission and denitrifying community. *Environmental Science & Technology*, *47*(13), 7341–7349. https://doi.org/10.1021/es305293h

Wang, C., Shen, J., Liu, J., Qin, H., Yuan, Q., Fan, F., Hu, Y., Wang, J., Wei, W., Li, Y., & Wu, J. (2019). Microbial mechanisms in the reduction of CH_4 emission from double rice cropping system amended by biochar: A four-year study. *Soil Biology and Biochemistry*, *135*, 251–263. https://doi.org/10.1016/j.soilbio.2019.05.012

Wang, C., Wang, W., Sardans, J., Singla, A., Zeng, C., Lai, D. Y. F., & Peñuelas, J. (2019). Effects of steel slag and biochar amendments on CO_2, CH_4, and N_2O flux, and rice productivity in a subtropical Chinese paddy field. *Environmental Geochemistry and Health*, *41*(3), 1419–1431. https://doi.org/10.1007/s10653-018-0224-7

Wang, D., Jiang, P., Zhang, H., & Yuan, W. (2020). Biochar production and applications in agro and forestry systems: A review. *Science of the Total Environment*, *723*, 137775. https://doi.org/10.1016/j.scitotenv.2020.137775

Wang, J., & Wang, S. (2019). Preparation, modification and environmental application of biochar: A review. *Journal of Cleaner Production*, *227*, 1002–1022. https://doi.org/10.1016/j.jclepro.2019.04.282

Wang, L., Yang, K., Gao, C., & Zhu, L. (2020). Effect and mechanism of biochar on CO_2 and N_2O emissions under different nitrogen fertilization gradient from an acidic soil. *Science of the Total Environment*, *747*, 141265. https://doi.org/10.1016/j.scitotenv.2020.141265

Wang, Q., Awasthi, M. K., Ren, X., Zhao, J., Li, R., Wang, Z., Chen, H., Wang, M., & Zhang, Z. (2017). Comparison of biochar, zeolite and their mixture amendment for aiding organic matter transformation and nitrogen conservation during pig manure composting. *Bioresource Technology*, *245*, 300–308. https://doi.org/10.1016/j.biortech.2017.08.158

Wang, S., Ma, S., Shan, J., Xia, Y., Lin, J., & Yan, X. (2019). A 2-year study on the effect of biochar on methane and nitrous oxide emissions in an intensive rice–wheat cropping system. *Biochar*, *1*(2), 177–186. https://doi.org/10.1007/s42773-019-00011-8

Wang, X., Lu, P., Yang, P., & Ren, S. (2021). Effects of fertilizer and biochar applications on the relationship among soil moisture, temperature, and N_2O emissions in farmland. *PeerJ*, *9*, e11674. https://doi.org/10.7717/peerj.11674

Woolf, D., Amonette, J. E., Street-Perrott, F. A., Lehmann, J., & Joseph, S. (2010). Sustainable biochar to mitigate global climate change. *Nature Communications*, *1*(56), 1–9. https://doi.org/10.1038/ncomms1053

Wu, H., Lai, C., Zeng, G., Liang, J., Chen, J., Xu, J., Dai, J., Li, X., Liu, J., Chen, M., & Wan, J. (2017). The interactions of composting and biochar and their implications for soil amendment and pollution remediation: A review. *Critical Reviews in Biotechnology*, *37*(6), 754–764. https://doi.org/10.1080/07388551.2016.1232696

Wu, Z., Song, Y., Shen, H., Jiang, X., Li, B., & Xiong, Z. (2019). Biochar can mitigate methane emissions by improving methanotrophs for prolonged period in fertilized paddy soils. *Environmental Pollution*, *253*, 1038–1046. https://doi.org/10.1016/j.envpol.2019.07.073

Wu, Z., Zhang, X., Dong, Y., Li, B., & Xiong, Z. (2019). Biochar amendment reduced greenhouse gas intensities in the rice-wheat rotation system: Six-year field observation and meta-analysis. *Agricultural and Forest Meteorology*, *278*, 107625. https://doi.org/10.1016/j.agrformet.2019.107625

Xiang, J., Liu, D., Ding, W., Yuan, J., & Lin, Y. (2015). Effects of biochar on nitrous oxide and nitric oxide emissions from paddy field during the wheat growth season. *Journal of Cleaner Production*, *104*, 52–58. https://doi.org/10.1016/j.jclepro.2014.12.038

Yang, W., Feng, G., Miles, D., Gao, L., Jia, Y., Li, C., & Qu, Z. (2020). Impact of biochar on greenhouse gas emissions and soil carbon sequestration in corn grown under drip irrigation with mulching. *Science of the Total Environment*, *729*, 138752. https://doi.org/10.1016/j.scitotenv.2020.138752

Yang, X., Lan, Y., Meng, J., Chen, W., Huang, Y., Cheng, X., He, T., Cao, T., Liu, Z., Jiang, L., & Gao, J. (2017). Effects of maize stover and its derived biochar on greenhouse gases emissions and C-budget of brown earth in northeast China. *Environmental Science and Pollution Research*, *24*(9), 8200–8209. https://doi.org/10.1007/s11356-017-8500-0

15372537, 2023, 4, Downloaded from https://acsess.onlinelibrary.wiley.com/doi/10.1002/jeq2.20475 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Yu, L., Tang, J., Zhang, R., Wu, Q., & Gong, M. (2013). Effects of biochar application on soil methane emission at different soil moisture levels. *Biology and Fertility of Soils*, *49*(2), 119–128. https://doi.org/10.1007/s00374-012-0703-4

Zhang, A., Bian, R., Hussain, Q., Li, L., Pan, G., Zheng, J., Zhang, X., & Zheng, J. (2013). Change in net global warming potential of a rice–wheat cropping system with biochar soil amendment in a rice paddy from China. *Agriculture, Ecosystems & Environment*, *173*, 37–45. https://doi.org/10.1016/j.agee.2013.04.001

Zhang, A., Bian, R., Pan, G., Cui, L., Hussain, Q., Li, L., Zheng, J., Zheng, J., Zhang, X., Han, X., & Yu, X. (2012). Effects of biochar amendment on soil quality, crop yield and greenhouse gas emission in a Chinese rice paddy: A field study of 2 consecutive rice growing cycles. *Field Crops Research*, *127*, 153–160. https://doi.org/10.1016/j.fcr.2011.11.020

Zhang, A., Cheng, G., Hussain, Q., Zhang, M., Feng, H., Dyck, M., Sun, B., Zhao, Y., Chen, H., Chen, J., & Wang, X. (2017). Contrasting effects of straw and straw–derived biochar application on net global warming potential in the Loess Plateau of China. *Field Crops Research*, *205*, 45–54. https://doi.org/10.1016/j.fcr.2017.02.006

Zhang, A., Cui, L., Pan, G., Li, L., Hussain, Q., Zhang, X., Zheng, J., & Crowley, D. (2010). Effect of biochar amendment on yield and methane and nitrous oxide emissions from a rice paddy from Tai Lake plain, China. *Agriculture, Ecosystems & Environment*, *139*(4), 469–475. https://doi.org/10.1016/j.agee.2010.09.003

Zhang, A., Liu, Y., Pan, G., Hussain, Q., Li, L., Zheng, J., & Zhang, X. (2012). Effect of biochar amendment on maize yield and greenhouse gas emissions from a soil organic carbon poor calcareous loamy soil from Central China Plain. *Plant and Soil*, *351*(1), 263–275. https://doi.org/10.1007/s11104-011-0957-x

Zhang, C., Zeng, G., Huang, D., Lai, C., Chen, M., Cheng, M., Tang, W., Tang, L., Dong, H., Huang, B., & Wang, R. (2019). Biochar for environmental management: Mitigating greenhouse gas emissions, contaminant treatment, and potential negative impacts. *Chemical Engineering Journal*, *373*, 902–922. https://doi.org/10.1016/j.cej.2019.05.139

Zhang, Q., Xiao, J., Xue, J., & Zhang, L. (2020). Quantifying the effects of biochar application on greenhouse gas emissions from agricultural soils: A global meta-analysis. *Sustainability*, *12*(8), 3436. https://doi.org/10.3390/su12083436

Zhang, Y., Zhang, Z., & Chen, Y. (2021). Biochar mitigates N_2O emission of microbial denitrification through modulating carbon metabolism and allocation of reducing power. *Journal Soil Water Conservation*, *55*, 8068–8078. https://doi.org/10.1021/acs.est.1c01976

Zhao, Q., Wang, Y., Xu, Z., & Yu, Z. (2021). How does biochar amendment affect soil methane oxidation? A review. *Journal of Soils and Sediments*, *21*, 1575–1586. https://doi.org/10.1007/s11368-021-02889-z

Zhou, H., Zhang, D., Wang, P., Liu, X., Cheng, K., Li, L., Zheng, J., Zhang, X., Zheng, J., Crowley, D., & Pan, G. (2017). Changes in microbial biomass and the metabolic quotient with biochar addition to agricultural soils: A meta-analysis. *Agriculture, Ecosystems & Environment*, *239*, 80–89. https://doi.org/10.1016/j.agee.2017.01.006

Zimmerman, A. R., Gao, B., & Ahn, M. Y. (2011). Positive and negative carbon mineralization priming effects among a variety of biochar-amended soils. *Soil Biology and Biochemistry*, *43*(6), 1169–1179. https://doi.org/10.1016/j.soilbio.2011.02.005

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